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Designing Collaboration Infrastructure in Challenge-Driven Labs: A Service Design Perspective

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Abstract

Universities are increasingly hosting challenge-driven education that bring together students, faculty, and external organizations to address complex societal problems. While the potential pedagogical benefits of these initiatives are well documented, the quality of the collaboration experience for the actors involved remains inconsistent, and existing research has largely approached the issue from a pedagogical rather than a process-oriented perspective.

This thesis investigates how stakeholder collaboration is experienced and coordinated in university-led challenge-driven educational context and proposes improvements to the underlying infrastructure from a service design perspective. Two cases at KTH Royal Institute of Technology are studied: the Global Development Lab (GDL), an emerging cross-institutional initiative, and MF2088 Innovation and Product Development, an established challenge-driven course. Data were collected through eight semi-structured interviews with students, coordinators, supervisors, and external partners, and analyzed using thematic analysis combined with service design tools, including stakeholder mapping, journey mapping, and service blueprinting.

Based in an analysis of stakeholder´s different experiences of participating in a challenge-driven education, the study identifies five systemic tensions that shape the collaboration across its lifecycle: temporal misalignment, asymmetric engagement, underbuilt support, under-designed backstage support, and ambiguous mutual value. These tensions are interpreted as infrastructural rather than interpersonal or pedagogical conditions. In response, the thesis proposes four service design principles and a conceptual intervention, the Collaboration Canvas, that translates the principles as a shared coordination layer across stakeholders. The study contributes to research on challenge-driven education by bringing in a process-oriented perspective using service design research and practice and offers a starting point for redesigning the collaborative infrastructure of future challenge-driven education and labs.

Keywords

Service design; Challenge-driven education; Stakeholder collaboration; Global Development Lab; University-society collaboration; Service blueprint

Sammanfattning

Universitet bedriver i allt högre grad utmaningsdrivna utbildning där studenter, lärare och externa organisationer samlas för att gemensamt hantera komplexa samhällsutmaningar. Även om de potentiella pedagogiska fördelarna med dessa initiativ är väl dokumenterade, visar studier att de involverade aktörernas upplevelser av kvaliteten i samarbetet skiljer sig åt. Befintlig forskning har i huvudsak närmat sig frågan ur ett pedagogiskt snarare än ur ett processorienterat perspektiv.

Denna uppsats undersöker hur samarbetet mellan intressenter upplevs och koordineras i universitetsledda utmaningsdrivna utbildningssammanhang och föreslår förbättringar av den underliggande infrastrukturen med stöd av ett tjänstedesignperspektiv. Två fall vid Kungliga Tekniska högskolan (KTH) studeras: Global Development Lab (GDL), ett framväxande tvärinstitutionellt initiativ, samt MF2088 Innovation och produktframtagning, en etablerad utmaningsdriven kurs. Data samlades in genom åtta semistrukturerade intervjuer med studenter, koordinatorer, handledare och externa partner, och analyserades med tematisk analys i kombination med tjänstedesignverktyg, däribland intressentkartläggning, kundresekartläggning och tjänstedesignblueprint.

Analysen identifierar med utgångspunkt i intressenternas skilda upplevelser av att delta i utmaningsdriven utbildning fem systemiska spänningar som präglar samarbetet över tid: temporal felinriktning, asymmetriskt engagemang, otillräcklig stödstruktur, underutvecklat bakomliggande stöd och otydligt ömsesidigt värde. Dessa spänningar tolkas som infrastrukturella snarare än interpersonella eller pedagogiska förhållanden. Som svar föreslår uppsatsen fyra principer med utgångspunkt från tjänstedesign samt en konceptuell intervention, Collaboration Canvas, som omsätter principerna till ett gemensamt koordineringslager mellan intressenter.

Studien bidrar till forskningen om utmaningsdriven utbildning genom att tillföra ett process-orienterat perspektiv med stöd av tjänstedesignforskning och praktik och erbjuder en utgångspunkt för att omforma samverkansinfrastrukturen i framtida laboratorier.

Nyckelord

Tjänstedesign; Utmaningsdriven utbildning; Intressentsamverkan; Global Development Lab; Universitet-samhälle-samverkan; Tjänstedesignblueprint

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List of acronyms and abbreviations

CBL	Challenge-Based Learning
CDE	Challenge-Driven Education
GDH	Global Development Hub
ECTS	European Credit Transfer and Accumulation System
GDH	Global Development Hub
GDL	Global Development Lab
GTL	Global Transformation Lab
KTH	KTH Royal Institute of Technology
MSD	Multilevel Service Design
PBL	Project-Based Learning
PPBL	Problem- and Project-Based Learning
RQ	Research Question
S-D logic	Service-Dominant Logic

1 Introduction

Nowadays universities have played a critical role in not only knowledge sharing and research but also emerge as a collaborative platform engaging directly with societal challenges and grand issues [1][2]. Against this background, an increasing number of collaborative labs initiated by higher educational institutions [3][4][5][6]. Besides, university program attaches great importance to introduce collaborative project into courses [7]. Those initiatives enable students, university faculty, and external organizations to jointly explore sustainable, societal problems, grand challenges and co-develop new ideas [4][8]. However, while the pedagogical ambitions of these initiatives are well articulated, the design and quality of the stakeholder experience in collaboration remains under expectation or partially unsatisfactory [9][10][11]. This thesis adopts a service design perspective to examine and improve the stakeholder collaboration experience in university-led labs for societal transformation.

1.1 Background

In the face of increasingly complex societal and environmental challenges, higher education institutions are facilitating the transformation of pedagogical models to promote innovation and active societal engagement [1]. At the forefront of this shift is Challenge-Driven Education (CDE) [2], an approach where students, researchers, and external stakeholders collaborate on exploring and resolving real-world problems. In Sweden, institutions such as the KTH Royal Institute of Technology have played a pioneering role in this domain. KTH's Global Development Hub (GDH) has driven the pedagogical shift toward what is recognized in current academic discourse as "Global Transformation Labs" (GTLs) or Global Development Labs (GDLs), aiming to address global-local sustainability tensions through systemic change by bringing together transdisciplinary student team, African partner universities, industry representatives, and public sector actors with the aim of addressing global-local sustainability tensions [4]. Moreover, established master's project-oriented courses at KTH such as MF2088 Innovation and Product Development follow a similar structure, but build connection between students and one external partners (e.g., startups, manufacturing industries) for semester-long projects centered on a real-life challenge using a learn-by-doing philosophy [8].

The collaboration in these labs and courses is expected to provide mutual benefit: students gain practical experience engaging in real problem, and external partners benefit from fresh perspectives, rapid prototyping of concepts, and strategic reflection on complex challenges [2][4]. Beyond promising expectation, executing this multi-actor collaboration in a university setting requires a carefully designed process and flexible system of interaction, one that takes full responsibility for the divergent expectations, timelines, deliveries and dynamic capacities of participating actors.

To meet these expectations, the collaboration infrastructure should systematically include how stakeholders are recruited, onboarded, supported, and followed up with. And the whole process can be regarded or conceptualized as a service. Service design offers a set of analytical and developmental tools, including stakeholder mapping, journey mapping, and service blueprinting, etc. These tools assisted as a systematic framework to investigate how different actors experience this service across its full lifecycle [12][13]. Applying this approach makes it possible to not just focus on general descriptions of collaboration experience challenges, but instead extract specific touchpoints, pain points, and moments of value creation or breakdown from the whole journey during the whole process[14][15][16]. Within the collaboration ecosystem, different stakeholders' perspectives are analyzed together and complement one another to create an overall map. This

perspective is particularly timely given that the GDL's vision of continuous testing, evaluation, and redesign of its participatory architecture and methods and contribute to its final objectives [4].

1.2 Problem definition

Existing research on challenge-driven education demonstrated that collaborative labs and project-based courses can empower students, strengthen their capacities to work with others from different disciplines. It enables universities to move toward more experimental forms of societal engagement, maximizing their potentials [2][6]. The GDL concept further envisions these environments as arenas for transformative learning among all participants, building sustainability mindsets and driving systemic change. These findings confirm the pedagogical potential of the approach.

At the same time, besides the positive visions, some significant tensions have been identifies in research. Teachers and organizers report considerable variation in how external partners participate, including differences in commitment levels, expectations, communication patterns, and perceived value of the collaboration [10][11][17]. Rosén et al. [13] noted that the role of and practical interaction with external stakeholders are crucial but complex. Prior experiences from GTL pilots and the MF2088 course indicate that some organizations are highly engaged and see clear benefits from co-creation, while others have limited or fragmented involvement over the project lifecycle.

However, existing research has largely recognized and examined these tensions from a pedagogical perspective, providing theoretical solutions on what could students learn and focus, how teachers or organizer create better environment, and learning canvas that could be directly utilized during courses [3][11][18]. What remains absent is a process-oriented analysis of how the collaboration itself is structured and experience of different participants could be improved practically. No study has applied service design methods or draw on a service perspective, using tools such as journey mapping and service blueprinting, to systematically diagnose where and why the collaboration process succeeds or breaks down across the full lifecycle in university-led labs or courses. This thesis fills the gap.

1.3 Purpose and Research Questions

The purpose of this thesis is to explore how the collaboration between students, external organizations, and university coordinators unfolds within KTH's Global Development Labs (GDLs), and the MF2088 Innovation and Product Development course. Based on this understanding, the study proposes structured improvements to this collaborative infrastructure from a service design perspective. To achieve this, the following research questions direct the study:

RQ1. How do different stakeholders, including students, external organizations, and course coordinators, experience the collaboration process over time in challenge-driven education?

RQ2. What are the critical pain points, breakdowns, and unmet needs in the collaboration among different actors, and at which touchpoints do they occur?

RQ3. How can the collaboration infrastructure in challenge-driven education be redesigned, informed by service design principles, to better support meaningful and balanced stakeholder engagement?

The study contributes to the long-term development of challenge-driven education, and particularly the GDL concept at KTH, meeting its vision of iterative evaluation through redesign of the participatory architecture [4]. By strengthening the collaboration process between academia and

external stakeholders around complex societal challenges, the findings also own broader goals of improving knowledge exchange process and innovation for sustainable development.

1.4 Delimitations

This study is delimited to the context of KTH's Global Transformation Labs and the project-based course MF2088 Innovation and Product Development. It does not intend to review all kinds of challenge-based learning across higher education, nor does it try to evaluate pedagogic outcomes from them. The emphasis will be on the collaborative processes experienced by those participating in this challenge-based activity, analyzing from the perspective of service design.

Methodologically, the research relies on qualitative methods with a limited sample of stakeholder interviews rather than large-scale quantitative surveys. The proposed improvements focus on the service and systemic dimensions of collaboration. Although there may be suggestions for conceptual designs, technical developments such as programming or coding are beyond the scope of this thesis.

1.5 Terminology Clarification

In current academic discourse[4], the main teaching model and future goals are defined as "Global Transformation Labs" (GTL). The term *transformation* is explicitly used to emphasize a shift toward systemic change, deep societal transitions, and reciprocal knowledge co-creation. However, in practice, these activities are coordinated under KTH Global Development Hub (GDH). As a result, to align with existing administrative structures, the actual spaces and courses are officially called "Global Development Labs" (GDL). To maintain conceptual precision, this thesis establishes a clear distinction: GTL is employed when discussing the broader theoretical ambitions, pedagogical frameworks, and systemic design, whereas GDL describes the actual programs and physical spaces investigated here.

1.6 Structure of the thesis

The thesis follows a logical research sequence. Chapter 2 presents the theoretical framework, including challenge-driven education, university-society collaboration, stakeholder engagement, and service design. Chapter 3 describes the case context, including the GTL initiative at KTH and the MF2088 course. Chapter 4 elaborates the methodology, including data collection, thematic analysis, the use of service design tools, and ethical considerations. Chapter 5 presents the empirical results and analysis of the current collaboration experience and service journey. Chapter 6 formulates insights and conceptual improvements informed by the findings. Chapter 7 contains a comprehensive discussion of the results in relation to the theoretical framework and research questions. The final chapter provides a summary, reflects on the limitations of the study, and outlines directions for future development.

2 Theoretical Framework

This chapter sets up the theoretical foundations for studying stakeholder experience in university-led labs for societal transformation. It unfolds in four steps. Section 2.1 defines the concept of challenge-driven education (CDE) and the Global Transformation Lab (GTL). Section 2.2 reframes the labs as service ecosystems by introducing Service-Dominant logic and elaborates the concept of university-society value co-creation within the ecosystems. In section 2.3, systemic design is introduced as a practical approach to help understand the complex nature of these ecosystems. Section 2.4 shows how specific service design tools are used to put this analytical perspective into practice. Section 2.5 provides an overview of the framework and highlights the research gap this thesis aims to fill.

2.1 Challenge-Driven Education and the Global Transformation Lab Concept

2.1.1 Challenge-Driven Education

Challenge-driven education (CDE) refers to a set of learning approaches where students collaborate across disciplinary boundaries, interact with external stakeholders, and work on open-ended projects addressing complex real-world challenges, which also often related to sustainable development [6][17]. The term shares many common features with, and is sometimes used interchangeably with, challenge-based learning (CBL), problem- and project-based learning (PBL/PPBL). Besides, it is closely related to pedagogies such as experiential learning, service learning, and inquiry-based learning [19][20]. Gallagher and Savage [21] and Leijon et al. [22] provide systematic reviews on this landscape, pointing out that definitional clarity remains limited, with each label being interpreted and implemented in different and sometimes contradictory ways.

Despite this, there are still several common characteristics that can be identified across CDE implementations. Van den Beemt et al. [23] proposed the "CBL compass" as a framework to conceptualize this variety, identifying features such as student-centeredness, action orientation, large-scale project formats (typically 15–22.5 ECTS credits), and collaboration with external stakeholders. Rosén et al. [13] practiced these characteristics through a research study involving teachers from four CDE courses at three Nordic universities. This study ended up finding that all courses are mainly student-centered, project-based, and engage external partners who provide real-world problem contexts.

A critical tension in CDE concerns the nature and framing of the challenge itself. Teachers in Rosén et al. [13] agreed that challenges should be "wicked" enough to develop students' abilities to manage complexity and uncertainty, while remaining bounded to be explored within available time, resources and capabilities of a course. If a challenge is framed too broadly, students may struggle to define a feasible direction or make meaningful progress, reducing its educational value. Identifying and framing challenges through pre-studies, dialogues, and negotiations between students, teachers, and external stakeholders varies significantly both between courses and between projects within the same course. The variability in challenge framing and stakeholder involvement is directly relevant to this thesis, as it suggests that the quality of collaboration cannot be standardized but must be designed and supported through flexible infrastructure [13]. The research of Rosén et al. further introduced the term *transformation-driving education* to more clearly illustrate the potential of CDE as a learning approach to drive sustainability transformations both in and through education. This framing goes beyond CDE simply as a teaching method and highlighting the role of it as a key factor in promoting societal change, which is central to the GTL concept.

2.1.2 The Global Transformation Lab (GTL)

GTLs are defined as arenas for collaborative experimentation on transformations within and through education, research, and innovation [9]. The purpose of them is to bring together students, teachers, researchers, and representatives from industry, public sector, and civil society, to jointly identify grand issues on global challenges and co-create knowledge, ideas, interventions, and solutions.

The GTL concept is theoretically grounded in the concept of robust action that articulated by Ferraro et al. [24]. This framework suggests that transformative systemic change may improve through gradual development, as long as three connected strategies: participatory architecture, multivocal inscription, and distributed experimentation, are connected over time.

Participatory architecture refers to arrangements that bring different stakeholders together and maintain their engagement over time. In the GTL context, this builds on the long-term partnership with four universities, several industrial partners and is intended to expand through additional academic and societal partners. Multivocal inscription refers to the principles, policies, and methods that guide the work and support shared understanding across actors and settings, including references such as the EU GreenComp framework [25] and the Earth for All report [26]. Distributed experimentation refers to concrete projects where participants address global challenges while developing their capacity for future problem solving. As shown in the GTL white paper, activities are organized into three phases: a think tank phase for co-exploring of the focus area, transformation-driving projects that include student and research projects, and post-project actions where the project outcomes may be carried forward into innovation, deeper research, or societal engagement. The first GTL prototype was tested at KTH in 2024 with three Stockholm-based industry partners and students from several institutions [4].

2.1.3 Identified Challenges in CDE and GTL Collaboration

Despite positive provision of the value delivered by CDE-type collaboration, there are some tensions identified by existing research. Teachers use "clients," "challenge-providers," and "challenge-owners" to refer to key external stakeholders [13], but the ambiguity in these role definitions contributes to misaligned expectations. Gulikers et al. [11] conducted an empirical research on transdisciplinary learning environments and found that external stakeholders were frequently reported to have varying levels of engagement and temporal misalignment across different phases of the co-creation process. Abegglen et al. [10] further highlighted that partnering with external stakeholders in higher education involves navigating differences in commitment, communication, and perceived value in the process of collaboration.

Based on the GTL's pilot experience, students reported that they highly valued the opportunity to broaden their understanding of society and future thinking through collaboration with industries. But meanwhile teachers observed clear limitations in both students' and external stakeholders' knowledge and methods regarding governance in complex contexts [9]. These findings suggest that collaboration quality is not only affected by isolated operational issues or individual actors' capabilities, but by the interaction of multiple stakeholders, institutional arrangements and coordination processes across the collaboration. Therefore, the collaboration breakdowns should be understood as systemic challenges emerging from the design of the collaboration ecosystem, rather than as isolated failures of individual participants. This perspective motivates the analytical approach introduced in the next section.

2.2 University-Society Collaboration and Innovation Ecosystem

Traditionally, university-led collaborative initiatives such as the CDE courses and GTLs have primarily been studied from an educational perspective, focusing on learning outcomes, the development of student competencies, and curriculum design [27], [28]. While this perspective is critical, it has limited analytical capacity to analyze the collaborative experiences of non-student stakeholders such as industry partners and public institution. As within the innovation ecosystem, no single stakeholder operates in isolation, the effective interaction of them is essential for the ecosystem's functionality and innovation outcomes[29]. And in the GDL context, the reason of participation, forms of engagement and timelines of external stakeholders sometimes largely differ from those of students.

Therefore, this research introduces a service-oriented perspective to complement the pedagogical perspective. The shift in perspective is grounded in the observation that GTLs are explicitly designed as collaborative infrastructures that simultaneously support multiple forms of value creation: student learning, societal impact, organizational innovation for external partners, and knowledge production for the university [4]. This multi-actor, multi-value scenario is well-suited for analysis through a Service-Dominant (S-D) logic.

2.2.1 Service-Dominant Logic and Value Co-Creation

Service-Dominant (S-D) logic was initially proposed by Vargo and Lusch [30] and has been continuously refined [31][32][33], providing a theoretical alternative to the traditional goods-dominant framework to understand economic exchange and value creation. The core idea is that service, defined as the application of knowledge and skills for the benefit of another actor, is the fundamental basis of all exchange [30]. Value is not simply produced by one party and delivered to another, instead, it is always jointly created through the integration of resources by multiple actors operating within shared institutional arrangements [34].

Several foundational premises of S-D logic are particularly relevant to the GTL context. First, One of the foundational premises of S-D logic (FP9) states that "all social and economic actors are resource integrators" [31]. In GTL, this means that students, teachers, external stakeholders including industry partners and public organizations are not merely providers or consumers of a service, but all contribute resources such as knowledge, networks, problem contexts, funding and legitimacy to the collaborative process. Second, S-D logic has evolved from a dyadic view of exchange toward a service ecosystem perspective [32][34], in which value co-creation is understood as occurring within the networks of interconnected actors coordinated through shared institutions with the same norms, symbols, and practices. This macro-level perspective is crucial for understanding GTLs, where collaboration crosses multiple institutions, disciplines, and national contexts.

Furthermore, S-D logic emphasizes that value is experienced differently by different actors and is shaped by how they use and interpret a service within their own contexts[35]. In the GTL setting, the "value" of collaboration differs across each stakeholder group: students derive learning outcomes and professional development; external organizations gain access to novel perspectives, innovation capacity, and potential solutions; teachers and researchers contribute to educational development and knowledge production; and the university as an institution strengthens its societal engagement mission. Understanding these differentiated value propositions and the points where they conflict or misalign is central to diagnosing collaboration breakdowns.

Viewing GTLs through S-D perspective implies that the collaboration issues identified in the existing literature, such as fragmented stakeholder engagement, different level of participation across project phases, and unclear expectations [10][11][13], are not individual failures. Rather, they reflect systemic misalignments in how resources are integrated and value is co-created within the

service ecosystem. To examine and address such misalignments, an analytical approach capable of handling the complexity of multi-actor in the multi-level service ecosystems is essential. This is where system thinking and service design intersect.

2.3 Systems Thinking, Service Design and the Intersection

2.3.1 Systems Thinking and Complexity

Systems thinking provides a set of concepts and tools to understand phenomena as a whole system rather than as collections of components [36]. Meadows [36] defined a system as "an interconnected set of elements that is coherently organized in a way that achieves something", and emphasized that the behavior of complex systems is driven by their interconnections, feedback loops, and properties of the systems as a whole. These behaviors, however, cannot be understood by analyzing individual components in isolation.

In the context of CDE, a systems thinking perspective indicates that the observed variability in collaboration is unlikely to result from only single cause. Instead, it should arise from the interaction of multiple factors operating at different scales: institutional norms governing academic timelines, organizational cultures and objectives of external partners, power dynamics between students, professionals and lab leaders, and the inherent ambiguity of wicked problems. Building on this perspective, this study distinguishes between localized insights and systemic insights. Localized insights describe specific pain points, needs, or behavioral patterns identified from empirical data at particular touchpoints. Systemic insights, by contrast, synthesize multiple localized insights across stakeholders and stages of the collaboration process to reveal the underlying structural dynamics and interdependencies that generate recurring patterns of breakdown or value creation.

Systems thinking also introduces the concept of leverage points, stating the nodes that within a complex system where even a small intervention can produce significant changes in overall behavior [36]. Such leverage points are unlikely to be identified by examining isolated events or individual actors alone. Instead, they can be uncovered by systematically analyzing stakeholder experiences, recuring pain points, coordination breakdowns and unmet needs across the collaboration process. This perspective aligns closely with service design, which utilizes tools such as stakeholder mapping, journey mapping and service blueprinting to reveal where interventions can have the greatest systemic impact. Therefore, service design methods in this study are regarded and used as practical means of identifying leverage points within collaboration.

2.3.2 Service Design as a Practice Discipline

Service design is a human-centered, iterative approach to create and improve services, focusing on the quality of the user experience and the orchestration of people, processes, and technologies [12]. Stickdorn et al. [12] characterized service design by six core principles: human-centered, collaborative, iterative, sequential, real, and holistic. The book provides actionable methodologies, such as journey mapping, service blueprinting, and prototyping, which make the abstract concept of "service quality" tangible and designable.

Stakeholder mapping helps identify and visualize the composition of parties that integrate resources within a service ecosystem, thereby putting into practice the S-D logic that “all social and economic actors are resource integrators”. Customer journey mapping captures the longitudinal, emotional and practical experience of a focal stakeholder across touchpoints and is widely recognized as a core service design tool for revealing invisible pain points and breakdowns [37], [38]. The service blueprint, originally conceptualized by Shostack [39] and subsequently refined by Bitner, Ostrom, and Morgan [40], is a process visualization technique that maps the service delivery process. Service blueprints expand this perspective by simultaneously representing customer actions, frontstage

activities, backstage processes and supporting systems, especially highlighting how invisible organizational arrangements shape the visible service experience [40].

The Multilevel Service Design (MSD) proposed by Patrício et al.'s [41] further clarified how such tools can be integrated across various levels, providing a multilevel logical framework for analyzing complex service ecosystems. This multilevel logic provides the basis for the methodological choices of this thesis. Chapter 4 returns to these tools and details how they are adapted to study GTL collaborations at ecosystem, journey and encounter levels.

However, service design is not only a toolkit, but a mindset aimed at understanding and designing experience of all stakeholders involved in a service, including those whose roles are typically invisible and often overlooked, normally like backstage support processes. This orientation is directly relevant to GTL collaboration, where much of the effort that enables successful student-company interaction occurs behind the scenes, in the form of matchmaking, onboarding, expectation management, and conflict resolution by course coordinators and GDL staff.

2.3.3 Systemic Design: The Intersection

The organic integration of systems thinking and service design practice is increasingly referred to as systemic design. Jones and Van Ael defined systemic design as combining the analytical capacity of systems thinking with the generative capacity of design. The former refers to the ability to map interconnections, feedback loops, and emergent behaviors. The latter refers to the ability to envision and prototype alternative futures. Besides, Sevaldson [42] indicated that systemic design is fundamentally a design approach focus on systems, placing greater emphasis on visual thinking, stakeholder engagement, and iterative interventions rather than purely analytical modelling.

This natural integration addresses the inherent challenge in the context of studying multi-stakeholder collaborations such as GTLs: systems thinking alone can describe why collaboration is complex, but it does not provide any methods to navigate and intervene in the complexity. Service design alone can visualize invisible supports and improve specific touchpoints, but without a systemic perspective, it risks optimizing isolate components, lacking a big-picture perspective and ignoring structural interdependencies. Systemic design bridges the gap by providing both the analytical guidance to understand the GTL as a complex dynamic system and the methodological toolkit from service design to reshape its collaborative architecture.

Systemic design provides a coherent theoretical foundation for the methodology employed in the study: using service design tools not as ends in themselves, but rather as a means to reveal the structure of collaboration systems and identify leverage points for improvement.

2.4 Summary

This chapter has established the theoretical foundations for the study by bringing together four streams of literature, as summarized in Table 2-1.

Table 2-1 Summary of the Theoretical Framework

Theoretical Concept	Key Sources	Role in This Study
Challenge-driven education & GTL concept	Rosén et al. [4]; Gallagher & Savage [21]; Leijon et al. [22]; Ferraro et al. [24]	Define the empirical context and the identified collaboration challenges
Service-Dominant logic	Vargo & Lusch [30], [31], [34]; Vargo, Maglio & Akaka [35]	Provide the macro-level perspective for reframing GTL as a service ecosystem with multiple resource-integrating actors
Systemic design	Meadows [36]; Jones & Van Ael [43]; Sevaldson [42]; Stickdorn et al. [12]	Justifies the integration of analytical and generative methods for complex multi-stakeholder settings
Service design tools	Shostack [39]; Bitner et al. [40]; Patrício et al. [41]; Følstad & Kvale [37]	Operationalize the analytical framework through concrete instruments applied in the methodology

The key research gap addressed by this thesis lies at the intersection of these streams. While CDE and GTL literature has documented the existence of collaboration challenges and the variation in stakeholder engagement (Section 2.1), no prior study to our knowledge has applied a service design and systemic design perspective to systematically map, diagnose, and redesign the stakeholder experience in these settings. To clarify, this chapter addresses GTL research, while the following case analysis focuses on the practical GDL implementation. The study fills the gap by employing the concepts and tools introduced above to analyze the collaboration process, with the assistance of similar courses experience at KTH, and to develop actionable design concepts for future improvements.

3 Case Context

3.1 Global Development Labs at KTH and Partner Institution

The empirical setting for this thesis focuses on a cross-institutional initiative that has evolved over almost a decade. In 2015, KTH Royal Institute of Technology in Stockholm established a strategic partnership with Botho University (BU) in Botswana and Lesotho, Strathmore University (SU) in Kenya, and the University of Dar es Salaam (UDSM) in Tanzania[4]. This collaboration was formally structured in 2017 under the coordination of the KTH Global Development Hub (GDH). The core objective of this partnership is to develop new educational models that address global sustainability challenges in direct alignment with the UN 2030 Agenda [2], [4], [13].

Over the years, the network has grown into a trust-based community through joint development of challenge-driven education (CDE) approaches, faculty training programmes, student mobility exchanges, and recurring conferences [4].

Although the initiative was originally led by KTH, it has since developed into a reciprocal partnership where the partner institutions co-create both learning and agenda-setting [4]. Physical hubs have also been opened at partner sites, for instance the Y4C Innovation Hub at UDSM and the Maggan Hub at KTH. Each function as a base for challenge-driven activities [4], [44].

Building on this foundation, the concept of Global Transformation Labs (GTL) has recently emerged as a further evolution of the partnership's activities [4]. The GTL concept is viewed as a platform where students, teachers, researchers, and representatives from industry, the public sector, and civil society jointly identify deep questions on global challenges and co-create knowledge, ideas, and interventions [4]. The internal structure of a GTL is organized around three interrelated activity streams: (1) a "Think Tank" where diverse actors co-explore the focus area and formulate societal challenges, project ideas, and research questions; (2) "Transformation-driving projects" at the core, encompassing challenge-driven student courses, thesis projects, and research projects; and (3) "Post-project actions" through which learning outcomes and project outputs feed into innovation, deeper research, or change agency [4]. These core activities are surrounded by seminars, workshops, trainings, and informal "hanging around" moments. Together they aim to create openness and cross-pollination among participants [4].

The first GTL prototype was initiated by the KTH Global Development Hub and piloted in 2024, involving KTH teachers, three Stockholm-based industry and institute partners, and students from KTH, Strathmore, Botho, as well as other international exchange students at KTH [4]. Initial feedback was overwhelmingly positive. Industry partners appreciated how the program expanded their employees' perspectives on future societal trends. Students found the approach much more integrated than traditional classes. Instructors highlighted the creative use of physical space, though they also pointed out a gap in participants' understanding of complex governance issues [4]. A second round of prototyping continued during 2025, with ongoing dialogue among the partner universities on how to develop joint GTLs across institutions [4] [45].

This thesis focuses on a single empirical case the Re-imagine Mobility Lab. The Lab is institutionally framed by the GDL, which provides leadership, administrative support. Within this frame, two parallel student project tracks operate alongside each other and engage the same external partner. The first is a voluntary project track, where the student coordinator recruits and organizes project-based students. The second is a course-based track from *DM2803 Transformative Change in Complex Systems, Project Course (15.0 credits)*, a mandatory course within the Master's Programme in Sustainable Digitalization at KTH [46]. Although both tracks share the same external partner, they differ in their internal stakeholder configurations. The course track involves formal supervision, examination, and lecturing by the course teachers, while the voluntary track relies on

the supervising and organizing role of a student coordinator. This setting forms the primary empirical case of the thesis.

3.2 Related Challenge-Driven Course

In addition to the GDL itself, this thesis also investigates another related challenge-driven course at KTH that share a similar pedagogical orientation but differ in course structure, disciplinary roots, and stakeholder engagement models. Studying this course alongside the GDL enables a comparative perspective on how different implementations of university-society collaboration shape the stakeholder experience. The two case contexts were selected based on supervisory recommendations that highlighted the value of comparing courses with different approaches to project-based and problem-based learning.

3.2.1 MF2088 Innovation and Product Development

MF2088 Innovation and Product Development is a 22.5-ECTS course offered at KTH within the MSc programme Integrated Product Design, the Innovation Management and Product Development track [4], [13]. The course enrolls primarily Swedish and international students from the Integrated Product Design programme, as well as occasionally students from KTH's GDH partner universities [4].

In MF2088, student teams work on real innovation challenges, typically in collaboration with manufacturing companies or, in some cases, public institutions [2], [4]. The course runs over one full semester and requires students to design their own innovation process using documented methods such as design thinking, lean start-up, and triple-layered business model canvas. Deliverables include business models, business plans, and solution concepts in the form of prototypes [4].

As documented in Rosén et al. [4] , MF2088 has its disciplinary roots in mechanical engineering but has gradually evolved towards more complex, open-ended projects where neither the problem nor the context can be fully specified in advance. This evolution has generated both pedagogical opportunities and institutional tensions, as the course's increasingly interdisciplinary and societally oriented character has raised questions of legitimacy within its home department [13]. The typical external stakeholders are manufacturing industry partners, though sometimes students develop their own project ideas [13].

From the perspective of this thesis, MF2088 represents a well-established, course-based model of university-industry collaboration with relatively structured stakeholder relationships. Unlike the GDL, where the collaboration is framed around global societal challenges and involves multiple partner institutions, MF2088 focuses on industry-defined innovation problems within a single institutional context. This difference provides a basis for comparing how challenge nature and stakeholder structures influence collaboration quality. Moreover, the two cases show similar patterns of collaboration and interaction among different actors, enabling a comparative analysis of stakeholder interactions in a comparable context.

3.3 Current Collaboration Setup Between Student and External Organizations

Across the two case contexts, the collaboration between universities and external organizations follows similar phases, though with different structural differences. This section synthesizes what is known from published sources and the thesis planning documents about the current collaboration setup. It will be further elaborated in Chapter 5 based on empirical interview data.

Initiation and partner recruitment

In the GDL context, external partners are typically recruited through the KTH Global Development Hub's existing network, relying on relationships built over nearly a decade of collaboration [4]. The process is largely relationship-based rather than standardized, with GDH coordinators brokering initial contact. In MF2088, external companies are recruited through a mix of channels: the course coordinator's industry network, returning partners from previous years, and occasionally student-initiated projects [13].

Onboarding and expectation alignment

Collaborative projects require aligning academic and industry expectations from the start. However, prior research and course documentation indicate that specific onboarding practices vary from case to case. In MF2088, the official course syllabus and memos do not explicitly prescribe a standardized onboarding protocol for external partners, meaning the level of formality depends on the specific partner [8]. Furthermore, historical course evaluations highlight that establishing clear expectations, roles, and activities between student teams and project partners during the initial phases remains a practical challenge [47]. Similarly, in the GDL context, while introductory meetings and challenge-scoping sessions take place, the specific onboarding format is still considered to be under development [4]. Consequently, understanding how expectations are aligned between academic requirements and organizational goals before project start is a central concern of this thesis and will be explored in depth through the empirical investigation.

Project execution and interaction patterns

During the active phase of the projects, student teams are expected to maintain regular contact with their external partners. According to the course guidelines, this collaboration typically involves scheduled milestone presentations and ongoing project meetings [47]. However, there is no single standard for how often or how closely they work together. The frequency, formality, and actual depth of these interactions can look completely different from one project to the next, even within the same course. This variation is especially common in setups with multiple parallel projects. As Rosén et al. [13] point out regarding MF2088, the student experience depends heavily on the specific company they are paired with. A partner’s daily availability, internal resources, and level of engagement can significantly change how the project is executed. Since these daily routines rely more on individual partner habits than on strict course rules, finding out how these communication patterns work in real life is another key area we will explore in our study.

Roles and power dynamics

External partner involvement varies, ranging from traditional clients who simply evaluate outputs to active participants who co-create with students. However, prior documentation in MF2088 suggests that role clarity is frequently lacking, leading to misaligned expectations between partners and students [47]. Similarly, while the GDL framework aims to establish a true partnership, existing literature points out that managing how power and responsibilities are distributed in these situations is still a significant challenge [4]. Understanding how partners and students agree on their roles during the project is another key focus of the research that follows.

Closure and follow-up

Collaborations typically conclude with a final presentation, but post-project outcomes vary widely. In MF2088, while some industry partners adopt student-developed concepts, others only view the collaboration as a low-commitment exploration [47]. Similarly, in the GDL context, the aspiration for 'post-project action' (such as continued innovation and change agency) is built into the conceptual framework but remains systematically unevaluated [4]. Since post-project outcomes lack formal documentation in both cases, our upcoming empirical analysis will investigate how closure is managed in practice.

Summary of case differences

Table 3-1 provides an overview comparison of the two case contexts with key dimensions relevant to this thesis.

Table 3-1 Comparison of the two case contexts

Dimension	GDL (Global Development Lab)	MF2088
Institutional home	KTH Global Development Hub (cross-institutional)	KTH Dept. of Engineering Design
Course extent	Varies by project; not a single fixed-credit course	22.5 ECTS, one semester
Course status	Experimental lab format, under iterative development	Established capstone course (mandatory in MSc Integrated Product Design)
Disciplinary roots	Transdisciplinary (no single disciplinary anchor)	Mechanical engineering
Student backgrounds	Students from KTH and GDH partner universities (Strathmore, Botho, UDSM); diverse disciplines	Primarily MSc Integrated Product Design students; mainly mechanical engineering background
Team size	Varies by project instance	5-7 students per team
Typical external stakeholders	Stockholm-based industries and institutes; public-sector actors in partner countries	Manufacturing industry; occasionally public institutions or student-initiated projects
Challenge type	Global societal challenges with local-global tensions	Industry-defined innovation problems; increasingly open-ended
Primary methodology	Emerging; guided by GTL framework (think tank, transformation-driving projects, post-project action)	Student-designed innovation process using design thinking, lean start-up, triple-layered business model canvas
Stakeholder recruitment	Relationship-based via GDH network; coordinator-brokered	Course coordinator's industry network; returning partners; occasionally student-initiated
Onboarding formalization	Introductory meetings and challenge-scoping sessions; format under development	No standardized protocol prescribed in course syllabus; varies by partner
Post-project follow-up	Conceptually built into GTL framework but not yet systematically evaluated	Varies; some partners adopt concepts; others treat collaboration as low-commitment exploration

4 Methodology

4.1 Research Process and Overall Design

4.1.1 Philosophical Stance and Research Strategy

This study adopts a qualitative research approach informed by a pragmatist stance. Pragmatism treats knowledge as inseparable from the context in which it is produced and values inquiry that leads to practical consequences [48]. This orientation was chosen for two specific reasons. First, the GTL concept itself is grounded in pragmatism. Rosén et al. [4] describe GTLs as arenas for evolutionary learning through continuous inquiry, reflection, and experimentation, drawing on Ferraro et al.'s [24] theory of robust action. This shared foundation aligns the research methodology directly with the practical nature of the collaborative labs. Furthermore, pragmatism justifies selecting methods based on their ability to answer specific research questions, rather than demanding strict adherence to a single philosophical tradition [49]. In this study, qualitative case study methods are used to understand stakeholder experiences in depth, while service design tools are used to translate those experiences into actionable improvement concepts. Pragmatism supports this methodology by prioritizing practical utility over strict paradigm adherence.

4.1.2 Qualitative Case Study as Research Design

This research uses a qualitative multiple-case study design. Case study research is suited to investigating complex social phenomena where the boundaries between the phenomenon and its real-world context are blurred [50]. Our study fits this model perfectly, since the collaboration processes, we are investigating cannot be meaningfully separated from the organizational and educational environments of their respective labs and courses.

We selected two cases at KTH for in-depth study. The first is the Global Development Lab (GDL) pilot. Launched in 2024 by the KTH Global Development Hub, local industries, and international students [4], the GDL is actively refining its organizational and pedagogical models. This allows the study to capture the emergence and evolution of collaboration dynamics in real time. The second case is MF2088 Innovation and Product Development, a challenge-driven capstone course with established industry ties. The course structure requires student teams to maintain deep collaborative ties with company representatives, teachers, and peers while addressing real industry challenges.

While both cases involve challenge-driven collaboration between students, staff, and external organizations, they differ significantly in maturity, structure, and scope. The GDL is a new initiative aimed at global societal transformation, while MF2088 is an established course focused on industry-specific product development. Comparing cases at different levels of maturity enhances the transferability of the findings. Recurring patterns observed across these diverse contexts further suggest underlying fundamental dynamics of university–industry collaboration [50].

The unit of analysis consists of the stakeholder collaboration experience, specifically the temporal sequence of touchpoints and relational exchanges between organizations, students, and faculty. A case study methodology provides the necessary depth and contextual focus [50]. This study applies a two-stage analytical framework. A within-case analysis to capture individual stakeholder contexts, followed by cross-case comparison to identify shared patterns and unique tensions [51]. This analysis yields a robust framework for informing the design of future challenge-driven educational and innovation platforms.

4.1.3 Integrating Service Design Methodology

In parallel with the case study, the research process is structured around the Double Diamond framework, which organizes design work into four phases of alternating divergent and convergent thinking: Discover, Define, Develop, and Deliver. This framework complements the case study by translating the analysis into actionable, human-centered outcomes [12].

In practice, the Discover and Define phases correspond to the empirical and analytical stages of the case study: semi-structured interviews with multiple stakeholder groups generate a broad account of collaboration experiences, which is then synthesized through thematic analysis into core themes and systemic bottlenecks. The Develop phase extends this analysis through service design artefacts, including stakeholder maps, journey maps, and service blueprints, which visualize the collaboration process and surface the relationships between frontstage interactions and backstage organizational activities.

The synthesis of case study methodology and service design ensures that the resulting knowledge is grounded in evidence while remaining directly applicable to the development of GDLs and similar initiatives.

4.2 Data Collection

4.2.1 Participants Selection

This research is conducted within the collaboration ecosystem containing different stakeholder which can mainly be categorized as three groups, the students, the university representatives and the industrial representatives. Under the context, the research adopted data triangulation to get multi-stakeholder data and study the overall journey that they all participated. In this way, the limitation inherently in relying on a single data source reduce can be largely reduced and thereby increase the ability to interpret and analyze them in the next phase[52].

Under the context of GDL and focus on researching collaboration experience of stakeholders, the research aimed to adopt purposive sampling[53], selecting those who had engaged in certain collaboration journey or truly participated in GDL context.

Two students, one teacher or coordinator, and one external partner were interviewed in each case. The participants and their projects are listed in Table 4-1.

Table 4-1 Interviewees

	Identity	Case	Project
1	Student coordinator	GDL	Voluntary project under Re-imagine mobility lab
2	Student 1		
3	Student 2		
4	External stakeholder 1		
5	Course teacher		Course project under Re-imagine mobility lab
6	Student 3	MF2088	Course project
7	Student 4		
8	Teacher supervisor		
9	External stakeholder 2		

GDL data is the main focus of this research, since the study is within the context of GDL and its goal of continuous improvement. MF2088 is used as a comparative case that provides additional student-company collaboration experience and shows how a similar collaboration structure works under a credit-bearing course context. Interview details and stakeholder roles are detailed in Chapter 5.

4.2.2 Semi-structured Interview

Data was collected primarily through different rounds of semi-structured interviews. Compared with other forms of interviews, it provides a flexible but guided framework, which ensures that the most critical topics regarding collaboration are covered while allowing participants' freedom to share any other experiences that they would like to talk about and sometimes unexpected insights exist[54].

A distinctive feature of the interview guides (Appendix A, B, C) is the integration of a service design perspective and journey-centric focus. Rather than asking generic questions about collaboration quality, the interviews were be structured chronologically around the lifecycle of the project they participated, which might include onboarding, regular meeting, execution point, and final delivery, etc. Participants were asked to describe specific touchpoints, communication flows, and their emotional highs and lows throughout the process. The whole interview guide is divided into different blocks following project timeline and each block is given a specific thematic coding number. It could be a great help in analysis and visualization phase since all the data have already attributed to certain themes and designed to be used in certain maps. This service-design-perspective questioning technique is purposefully designed to elicit the temporal and experiential data necessary for constructing accurate Journey Maps and Service Blueprints in the subsequent analysis phase.

4.3 Assessing Trustworthiness

This study adopts Lincoln and Guba's [55] trustworthiness framework in place of positivist validity and reliability criteria. The four criteria and the strategies used to address each are outlined below.

4.3.1 Credibility

Credibility is enhanced through data triangulation across three stakeholder groups (external organizations, students, and university staff) and two cases of different maturity levels (GDL and MF2088). This multi-source, multi-case design allows claims to be cross-checked and reduces the risk of one-sided accounts. Member checking is applied selectively by sharing draft findings and journey maps with willing participants, primarily lab organizers and course teachers, to verify the researchers' interpretations. Ongoing dialogue with the thesis supervisor, who is closely involved in the GTL initiative, provides an additional credibility check.

4.3.2 Transferability

Rather than seeking statistical generalization, the study pursues transferability through thick description of each case context, including organizational structures, participant roles, and project timelines. The inclusion of two cases with contrasting maturity levels helps distinguish context-specific findings from patterns that may apply more broadly to challenge-driven university-industry collaboration.

4.3.3 Dependability and Confirmability

A transparent audit trail documents the entire analytical chain from interview transcripts through thematic codes to service design artefacts. The interview guide, coding scheme, and mapping logic are included in the appendices. Regarding transcription, interviews were recorded and initially transcribed using Granola, an AI-assisted tool; both researchers then verified the transcripts against the original recordings to ensure accuracy. Key analytical decisions, such as the adoption of a hybrid abductive coding approach, were discussed with the thesis supervisor and documented at each stage.

4.3.4 Researcher Reflexivity

As KTH master's students with direct experience of the MF2088 course, the researchers perform a position that is neither fully insider nor fully outsider [56]. This familiarity facilitated access and contextual understanding but also carried risks of confirmation bias. To mitigate this, the two researchers regularly debriefed each other throughout data collection and analysis, cross-checking interpretations and challenging emerging assumptions. Where fieldwork realities required methodological adjustments, such as replacing a planned digital timeline mapping exercise with oral temporal prompting after the first interview, these decisions were documented as part of the audit trail.

4.4 Data Analysis

The qualitative data collected from interviews were analyzed utilizing a combination of thematic analysis and service design tools with multilevel service design mindset. This approach allowed for changing identification and categorization of stakeholder patterns and behavior into different themes. These themes were systematically being visualized and compared facilitate a rigorous scientific investigation of the dataset.

This study combines localized finding derived from empirical data with systemic insights that connects multiple insights across stakeholders, touchpoints, and process stages to examine the service process from a systems thinking perspective.

4.4.1 Thematic Analysis

The interview was analyzed using thematic analysis to identify recurrent patterns. Following Lochmiller's [57] description of thematic analysis as an approach that connects individual codes, intermediate categories and researcher-generated themes, the analysis combined deductive and inductive elements. A priori codes linked to the structure of the semi structured interview guide were used to organize the data, while higher order themes were developed through iterative comparison within and across cases.

The analysis followed a two-layer logic. The first layer consisted of deductive TC-X codes that were defined before data collection from the interview guide, covering domains such as touchpoints (TC-T), motivations and expectations (TC-M), value creation (TC-V), backstage support (TC-S), pain points (TC-P), roles (TC-R) and future needs (TC-F). These codes were used to segment the transcripts and to serve the journey map and service blueprint outputs. The second layer consisted of inductive themes that were built from these domains through cross-case comparison. The five systemic tensions reported in Section 5.4.1 (T1 to T5) belong to this second layer and were not predefined.

In practice, all data were transcribed and read thoroughly to check that the TC-X scheme had been applied consistently. Within each domain, the specific pain points, breakdowns, and stated needs raised by each respondent were noted. These were then compared across respondents and two cases.

The same underlying problem in different words were grouped together, and these recurring groups were consolidated into the five higher order tensions through discussion. Although seen linear, this process was iterative since the tensions were revisited and refined as further interviews were coded.

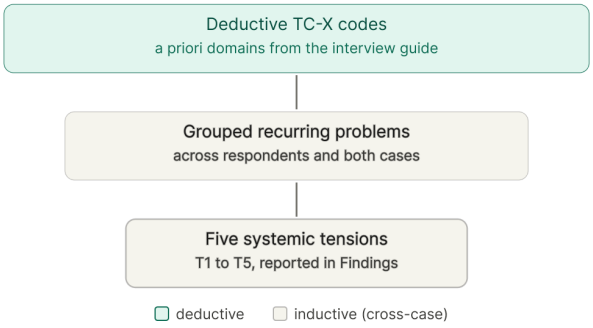


Figure 4.1 The thematic analysis process

The outcome of this process is reported in Chapter 5. The full list of the five tensions and their distribution across the nine participants and the two cases is shown in the tension distribution matrix in Figure 5.4. The matrix shows that each tension was raised by several participants and across more than one stakeholder group rather than reflecting a single account, which supports treating them as cross-case patterns. T5 was identified as a higher order tension that conditions T1 to T4 rather than as a parallel operational theme, as discussed in Section 5.4.1.

4.4.2 Service Design Tools and MSD

The study employs three foundational service design tools with the multilevel service design mentality. The service design tools are not only used as visualization techniques but as analytical instruments to identify recurring pain points, coordination breakdowns, and unmet stakeholder needs across the collaboration process. These recurring patterns are interpreted as potential systemic leverage points that informed the subsequent development of service design principles and concept proposals. Each tool operates at a different level of analysis, following the multilevel logic proposed by Patrício et al. [41].

Stakeholder Mapping

Stakeholder mapping is used to identify and visualize the actors, their roles, and the relationships between them within a service ecosystem [12]. In the GTL context, stakeholder mapping serves to make explicit the full constellation of actors involved (students, teaching staff, GDL coordinators, industry contact persons, company leadership, public organization representatives) and to reveal power distributions, information access, and level of involvement over time. This tool connects to the S-D logic perspective by operationalizing the concept of "resource-integrating actors" [31] at the ecosystem level.

Customer Journey Mapping

In this study, journey mapping is adapted from its typical commercial context to map the collaboration journey of external organizations and students through the GTL process, from initial contact and onboarding through active project collaboration to conclusion and follow-up. By capturing both functional activities and emotional states at each phase, journey maps enable the identification of critical moments where collaboration satisfaction rises or falls, and where and what kind of intervention may be most impactful.

Service Blueprint

Service blueprinting serves a diagnostic function: it reveals how backstage organizational processes, such as course coordination, company matchmaking, timeline synchronization, and resource allocation, either support or undermine the frontstage collaboration experience. This is particularly relevant to the GTL context, where prior research suggests that much of the variation in collaboration quality is attributable to backstage factors that external stakeholders are unaware of.

Multilevel Service Design

MSD enables integrated design at three hierarchical levels: (a) the service concept level, which maps the customer value constellation; (b) the service system level, which defines the service architecture and navigation; and (c) the service encounter level, which uses the Service Experience Blueprint to design individual interactions [41].

While the full MSD method is beyond the scope of this thesis, its multilevel logic informs the analytical structure. Specifically, the study examines those challenge-driven education at the ecosystem level (stakeholder mapping), the journey level (journey maps capturing the full collaboration process), and the encounter level (service blueprints diagnosing specific interaction points and backstage support). This multilevel approach ensures that the analysis captures both the macro-level systemic dynamics and the micro-level experiential details of stakeholder collaboration.

4.5 AI Usage Disclosure

While various AI-assisted tools were employed to support the process, full responsibility for the whole research remains with the authors.

To identify relevant literature, the Primo Research Assistant of KTH library was utilized, after which each source was manually assessed for quality and relevance before inclusion. Similarly, Granola was used to provide initial interview transcripts, but all contents were cross-checked against original audio recordings to ensure accuracy. Perplexity facilitated the discovery of additional references, which were also later verified through primary databases prior to citation.

Gemini and ChatGPT was used to refine and polish language, such as improving grammar and flow, without altering the underlying arguments. For the platform prototype, Claude served as a development assistant to build the canvas. No AI tool was used to generate data or replace critical analysis, and all final conclusions reflect the original work and judgement of the authors.

5 Current Collaboration System and Stakeholder Experience

The chapter analyses the collaboration in three steps. Section 5.1 maps the stakeholder ecosystem and how value is exchanged between actors. Section 5.2 using stakeholder journey maps to trace different phases of the projects. Section 5.3 highlights backstage support behind the journey using a service blueprint. Section 5.4 synthesizes these findings into five systemic tensions and opportunity areas, which form the basis of Chapter 6.

5.1 Collaboration ecosystem and stakeholder roles

5.1.1 The GDL stakeholder mapping

GDL was organized under GDH from KTH, operating by GDH staff as leadership or administrative. Re-Imagining Mobility Lab is the one conducted under GDL and utilizing all resources provided by the hub. There are two projects conducted in parallel under Re-Imagining Mobility Lab, one is an extra curriculum that run by selected student coordinators, the other is a program course in collaboration with the lab. Since the project conducted by either of them goes under Re-Imagining Mobility Lab context, they share same external partners, who were contacted by GDH leadership considering the lab visions. The course project officially belongs to a master program but have collaboration with GDH. Therefore, it has both administrative settings from course program similar to MF2088 and from GDL leadership.

Drawing on the voluntary project, the students are lab participants who join voluntarily, typically motivated by their interest in mobility and sustainability without constrain in credits. The student coordinator is the operational lead who runs the lab in practice, recruit students, designs sessions, manages outreach, and acts as the main internal point of contact, while also participating as a student. As for the course project, course teacher reuses the lab's challenge brief but defines its own learning outcomes, prerequisites, and pass-fail grading. The teacher also identifies as an educator, giving lecture on relevant principles or theories for both students from the program course and the voluntary project. The external partners give guest lectures, engage in the student project and give feedback. The course teacher does not own the partner contact directly, so communication with external stakeholder was through GDH, therefore alignment work between course expectations and partner expectations is mediated through GDH rather than negotiated face-to-face.

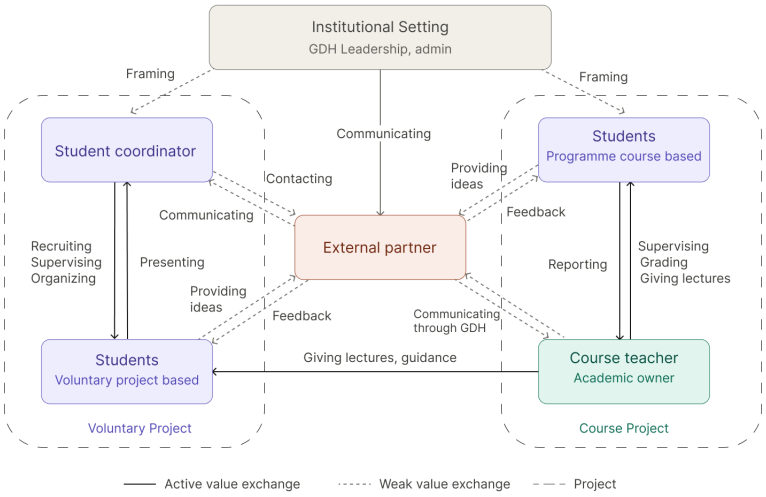


Figure 5.1 Re-imagining Mobility Lab stakeholder mapping

Within each project, there is a triangulation formed by students, coordinator/teacher and external partner, but course teacher also engages within the voluntary project as external knowledge providers. The course project, on the contrary, has similar role composition to MF2088.

5.1.2 The MF2088 stakeholder mapping

Leading by institutional settings, MF2088 operates as a triadic ecosystem consisting of students, teacher supervisor and external stakeholders. The students participating in the course are enrolled in master program. The project works as a mandatory course component with structured calendar and graded deliverables. The teacher supervisors are responsible for the course design, external partners recruitment, assessment criteria settings. Besides, they also play a role in giving introduction and onboarding external partners, as well as coaching students throughout the semester. The external partner contributes a concrete project, deeply engage within the student teams, providing prototyping access and direct feedback on student work.

The three actors are linked by direct working relations. Students and the partner exchange ideas directly. Teachers directly reach students through supervision, lectures and seminars. The communication between teachers and external stakeholders are less intensive with the educational objectives of capability of communication and coordination for students. The institutional setting, including course administration, funding rules, and fixed credit structures, reaches the teacher supervisor directly, who convert these constraints into project conditions communicated to the partner before the course starts.

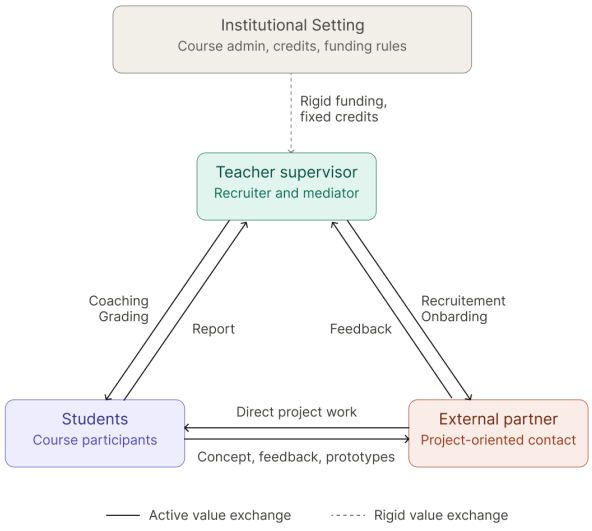


Figure 5.2 MF2088 stakeholder mapping

5.2 Collaboration Journeys Across the Project Lifecycle

The second level of analysis traces how different stakeholders collaborated over time. It follows four phases: initiation and recruitment, onboarding and alignment, project execution, and closure and continuity. Since the GDL data are all gathered from voluntary project participants, the following analysis explains only the voluntary project case.

5.2.1 Initiation and Recruitment

In GDL, initiation is the phase when asymmetries first occurred. For instance, the student coordinator reported that recent semesters began without a stakeholder-provided challenge, that the kickoff with stakeholders occurred late, and that they proceeded student recruitment with an undefined object of work. Students participated in the lab were attracted by their curiosity, sustainability commitment, and the vision of real-world cases engagement, but it turned out to be lacking defined scope to attach those motivations to. When the external partner was trying to find someone joining the collaboration, she found herself hard to answer her colleagues' question about why they should participate in the it and how the engagement would benefit the organization. The three actors arrive at initiation with diverse motivations, and no instrument exists at this phase to support a shared understanding.

In MF2088, initiation is supported by the course settings: instruction is provided, the schedule is fixed, and the recruitment of students is handled by the course support system rather than by individual coordinators. But on the other hand, the teacher supervisor reported that partner recruitment has become harder over time, since Swedish companies share less direct contact information than before, and that some partners take on the commitments that they cannot realistically meet. The MF2088 partner described running the project primarily as a project rather than through course context or aiming at achieving some learning goals, indicating that even when an academic framing exists, it does not reliably structure partner behavior. Even though the recruitment is difficult, and the collaboration is valuable to all actors, misalignment still existed. Because external partner may expect more beyond educational objectives and have different view of the project while students' perspectives are limited by the course scope.

5.2.2 Onboarding and Alignment

Onboarding is the phase when expectations should be aligned and collaboration should be anchored. In GDL, this phase is largely absent as a designed activity. Student 1 reported that there was no shared definition of why each party was present, as well as goals and intended outcomes were not clarified, and that companies appeared more "as logos" than as active contributors. The external GDL stakeholder reported difficulty identifying the right colleagues to involve, indicating that the partner side of the alignment was also under-supported. The student coordinator reported that even when feedback from stakeholders was received, it arrived too late and was not constructive enough to convert to realistic adjustment. Onboarding in GDL is not a constructive phase that all participants had been through, but an ignored phase whose absence gradually accumulates pressure on later phases.

MF2088 offers a more developed onboarding process through seminar arrangement, but the alignment achieved at this phase still remains fragile and under expectation. The teacher supervisor reported that anchoring is sometimes fragile because the initial company contact might probably transfer the project to a less committed person, resulting in anchor missing and mess in knowledge sharing. Student 4 reported that when the project proceeded to midterm, the partner proposed pursuing an unvalidated idea outside the alignment that established initially, indicating that what was aligned at onboarding phase might not be able to bind through execution. Evidence from MF2088 shows that the existence of an onboarding process is not sufficient, there is still a lack of anchoring mechanisms that help when personnel changes and misalignment occur.

5.2.3 Project Collaboration and Execution

Execution is the phase when collaboration intensively exists, and it is also the phase that respondents described in the greatest detail. In GDL, the execution phase is characterized by an

overly broad scope, and students are unable to narrow it down without external guidance. Student 2 reported that the team felt underqualified for the complexity of the challenge and a lack of expertise in the relevant fields prevented them from making confident decisions regarding the scope of the project. Although they have been told that the team was not required to provide a well-developed solution, the implicit pressure to satisfy the challenge giver still bring the student anxiety instead of freedom. Meanwhile, the external GDL stakeholder identified a tension between problems broad enough to support systems learning and problems narrow enough for organizational application. In this case, execution in GDL is a phase where scope, deliverables, and value are simultaneously under-defined, and the team faced a situation that they must make decisions even under this ambiguity.

In MF2088, Student 4 reported that partner engagement changed overtime and increased obviously late in the project, transforming the experience into something closer to work than coursework. External partner reported that the complexity in IT development forced them to frequently reprioritize functionality in solution. He also mentioned that moving from student concept to real-market testing was substantially harder than developing the concept internally. The teacher supervisor reported that end-user or customer access is often controlled by third parties that partner with them whose interests do not align with the project. Internally, team conflicts arose beyond teachers' expectation and sometimes was hard to resolve since they were not trained to mediate. The MF2088, friction existed in the collaboration between partner and its end users, partner and student team, and within student team, while in GDL, friction more existed at the center, where the work itself, the scope, how should it be done are under specified. But both patterns stem from the same issue: a lack of formalized roles and contributions, coupled with a persistent gap between academic and commercial goals.

5.2.4 Closure and Continuity

Closure is the critical phase distinguishing the lab from MF2088. In GDL, due to fragmented stakeholder feedback and unstandardized handover processes, closure did not meet the initial vision of the lab. Specifically, the student coordinator reported a rushed inter-semester transition that left students highly uncertain about how they were being evaluated. On the other hand, while external stakeholders expressed a broad willingness to continue collaboration, this intent was not grounded in tangible organizational outcomes or systematic feedback. As external stakeholder 1 stated that the outcome of the project is sometimes less meaningful for them than the exploring process. The realistic problems usually land in a complicated system which contains a lot more actors and typically the outcome provided by student team is more than difficult for only one actor to realize. Consequently, in the closure phase, the project was simply concluded without fostering institutional continuity.

As for MF2088, although it had a more formal closure with deliverables like final reports, prototype, business plan, and grading, its continuity remains similarly compromised. The supervisor highlighted that rigid funding rules actively discouraged iterative course redesign, preventing systemic insights from one practice from reshaping the next. Furthermore, external partner observed a similar implementation gap: student concepts were rarely transitioned into real-market adoption due to a distinct lack of post-project support for partner organizations. Across both cases, closure terminates the project but did not feed the system, and the absence of designed continuity mechanisms is itself a finding.

5.3 Service Blueprint

Service Blueprint of University-led Collaboration Lifecycle

Blueprint layer	1. Initiation & Recruitment	2. Onboarding & Alignment	3. Execution	4. Closure & Follow-up
Frontstage interactions *Visible touchpoints*	Visible entry points: partner invitation, project brief, kickoff workshop.	Shared framing moments: seminar, scope discussion, expectation setting.	Ongoing encounters: supervision, critiques, student–partner meetings.	Visible outputs: final presentation, report, assessment and feedback.
Frontstage frictions *Surface bottlenecks*	Briefs arrive late or remain too abstract for action.	Roles, value propositions and expected contribution remain uneven.	Scope changes, information gaps and team conflicts surface during work.	Handover and evaluation become rushed or ambiguous. line of visibility
Backstage actions *Invisible operational work*	Recruit partners; translate institutional ambitions into project opportunities.	Align partner motivation, student capacity and academic requirements.	Adjust timelines, mediate conflicts and maintain stakeholder relationships.	Collect feedback, balance workload and preserve continuity for future cycles.
Backstage vulnerabilities *Where individual burden appears*	Relies on individual initiative and weak institutional memory.	Fragile when contacts change or initial consensus is not anchored.	Coordination overload shifts systemic failures onto teachers/coordinators.	Post-project learning and continuity lack a stable owner. line of institutional support
Support processes *Institutional infrastructure*	Need: clearer value proposition and reusable recruitment materials.	Need: onboarding protocol, role agreement and shared collaboration timeline.	Need: conflict-mediation guidance, collaboration tools and access support.	Need: feedback loop, handover archive and conditions for iterative course redesign.
Meta-finding *Systemic interpretation*	**The collaboration is still managed as an administrative process rather than designed as a dynamic service system.**			

Figure 5.3 Simplified service blueprint of the current collaboration lifecycle

The third level of analysis applies a service blueprint framework to examine the lifecycle journey of the collaboration. The blueprint distinguishes frontstage interactions, backstage actions, and support processes to uncover invisible labor and structural dependencies. Figure 5.3 shows the blueprint, illustrating how the current support processes force backstage actors to individually absorb systemic failures, which ultimately manifests as recurring frontstage frictions.

The frontstage of the collaboration consists of visible touchpoints described readily by students and partners, such as workshops, meetings, and presentations. While GDL has a more open frontstage compared to the more intensive curriculum of MF2088, respondents in both cases reported frequent friction. This friction comes from events occurring outside the proper frontstage, such as late information arrival, rushed handovers, and unresolved team conflicts. The frontstage is what actors see, but it is not where the determinant work occurs.

The backstage layer is the operational core of the collaboration, but it receives the least institutional support. In GDL, the student coordinator manages stakeholder relationships, continuity design, and recruitment without sufficient leadership input. Similarly, in MF2088, the teacher supervisor absorbs equivalent backstage work, including recruiting partners under difficult conditions, mediating student conflicts without formal training, and manually adjusting workloads. Consequently, external partners find the course framework invisible to them during the engagement, indicating that the translation work the teacher performs is also incomplete from the partner perspective. Across both cases, the backstage functions as a zone where individual actors absorb systemic failures rather than a deliberately organized space.

Support processes form the institutional infrastructure such as training, funding, and knowledge management should enable backstage work. In GDL, this infrastructure is largely absent, lacking formalized handovers, onboarding procedures, or feedback mechanisms. In MF2088 and formal course project in GDL, the support layer is equally constrained under school administrative structure. Ultimately, the infrastructure in both cases treat the collaboration as a static administrative object rather than a dynamically designed service.

5.4 Key Pain Points, Needs, and Opportunity

This section synthesizes the empirical findings, extracting underlying tensions from the whole journey. These tensions reflect existing issues within collaboration that lead to breakdowns across different stakeholders, different phases within two cases. This kind of cross-case analysis is categorized in two distinct levels: T1–T4 summarize operational and infrastructural tensions that directly influence the collaboration performance, while T5 is a higher-layer tension regarding the overall objective and value of the collaboration. Figure 5.4 shows how the five tensions are distributed across the nine interview participants in the two cases.

5.4.1 Systemic Tensions

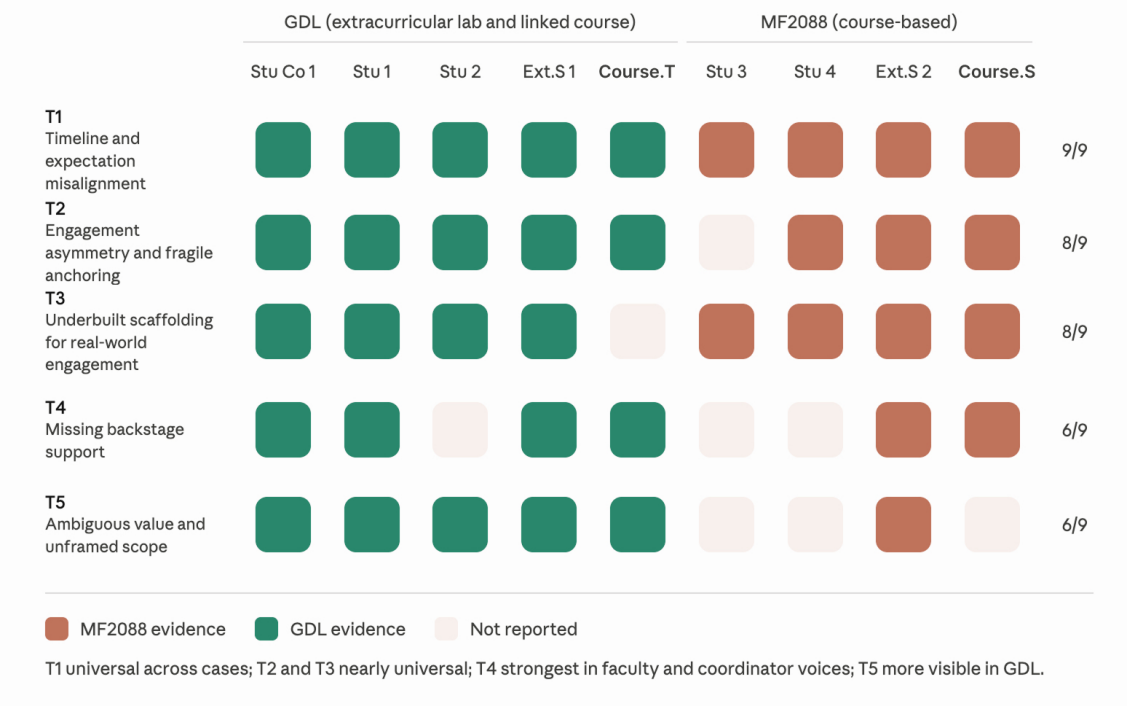


Figure 5.4 Cross-stakeholder pain point matrix across GDL and MF2088 respondents

T1. Temporal misalignment between academic and partner cycles

This tension stems from two incompatible operational rhythms that the current collaboration framework does not resolve. The academic schedules are restricted by semester deadlines, partner decision cycles follow external corporate priorities. In the GDL case, the student coordinator mentioned a late-February stakeholder meeting when they got disappointment about the previous semester. But it came too late to reshape the ongoing one. Also, the handover between semester was in a rush and the spring recruitment suffered from weaker orientation timing. Student 1 reported that partner decision cycles did not match the lab's pace, resulting in considerable delays in follow-up after meetings. The GDL external stakeholder added that expectations about company workload were unclear in both directions, time and effort sometimes underestimated and offered support sometimes remain unused. Besides, their prioritization on organizational work is inevitable.

In MF2088, similar dynamics appeared as a project timeline moving faster than the course timeline, late user access from partners, and emergent technical complexity that forced reprioritization. The pattern indicates a structural mismatch rather than a simply scheduling problem, since no individual actor himself can resolve the underlying rhythm difference.

T2. Asymmetric engagement and fragile anchoring

In the MF2088 and DM2083 courses, external partners engagement is mostly voluntary, while students are required to participate as a compulsory course to earn their credits. This basic imbalance causes continuous difficulties throughout the entire project. Within the GDL setup, Student 1 explained that some partners acted as symbolic participants rather than active team members, adding that because the lab was an extracurricular activity, partners felt less pressure to attend meetings. Student 2 noted that when partners failed to show up, the student team lost the external feedback needed to narrow down their project scope, so the project was temporarily unable to proceed. From the partner's perspective, external stakeholder 1 explained that it was hard to find colleagues with the right technical expertise to help the students. This stakeholder also observed that even though students technically had open access to them, the steep hierarchy between students and corporate professionals made students hesitant to reach out. Furthermore, the course teacher pointed out a similar gap in how relationships were managed, stating that the university "did not really own the contact with the stakeholders." Lacking a proper information sync, mismatches in expectations only became clear after the project ended. For instance, when a partner's suddenly expressed dissatisfaction, this "came as a surprise" because no issues had been raised through coordinators or previous communication. This gap is also visible from student coordinator, who reported that external partners did not provide any project challenges for the recent semester at early phase.

In MF2088, this imbalance led to practical disruptions. Partners demanded unverified changes to the project scope and sometimes conflicted with the academic schedule. Therefore, these ongoing conflicts happen because the current university framework only acts as a temporary middleman to introduce partners, instead of building a permanent system to secure their long-term commitment. The issue is rooted in this institutional setup.

T3. Underbuilt scaffolding for real-world engagement

Both cases aim to solve real-world problems, but neither provides the practical support needed to maintain these partnerships, such as clear project assignments, shared communication tools, or methods to help students and companies understand each other's goals. In the GDL case, this lack of scaffolding became clear at the very start of the project when defining the work. Student 1 and the student coordinator both noted that external partners failed to provide clear predefined project briefs from student's perspectives. Because of this, Student 2 explained that the topics remained too broad, leaving the student team confused about how to start working in an unfamiliar field. Adding to this communication gap, external stakeholder 1 observed that students' exploratory research and the company's commercial focus operated on completely different expectations, with no shared guidelines to align them.

In contrast, the MF2088 case suffered from a lack of support during the final execution and prototyping stage. Student 3 reported that there were no shared digital tools or platforms to collaborate with the company. In terms of output quality, Student 4 suggested that the course should prioritize the development of working prototypes instead of focusing mainly on conceptual proposals. However, limited access to time and prototyping facilities at the university makes it difficult to accommodate the varying requirements of different projects. Furthermore, the external MF2088 stakeholder highlighted how difficult it was for students to get in touch with actual customers for market testing. Across both cases, the core issue is not a lack of effort from the participants, but a lack of structured management from the course itself. The current setup fails to provide a consistent mechanism that connects the students' initial research to the practical outcomes that outside companies find valuable.

T4. Under-designed backstage and absent institutional adaptability

The daily management that keeps these projects running depends entirely on individual staff members rather than structured university support, and the school's administrative rules make it even difficult to fix problems when they arise. Within the GDL setup, the student coordinator explained that all instructions about the project were unclear no matter from university leadership or external partners. Furthermore, since the student coordinator changed every semester, the handover of information between semesters was poorly managed. Besides, the course teacher pointed out that no single person had a complete overview of the entire project. Information was scattered and communication power was owned by the organizer. He added that the administrative unit was overworked and lacked the staff to manage the program properly, which left external partners feeling disconnected and as a result their complaints went unnoticed until the project was already over. This lack of clear roles was also highlighted by the external GDL stakeholder, who pointed out that nobody was officially assigned to handle conflicts within student groups or to facilitate communication between students and companies.

In the MF2088 case, similar bureaucratic obstacles appeared in how the course itself was regulated. The teacher supervisor explained that rigid university funding rules made it difficult to revise the course design even with significant feedback. This supervisor also noted that has become harder for students to directly in contact with organization without the help of school and courses, because Swedish companies are increasingly reluctant to share direct corporate contact details with the public. In terms of administrative policy, the supervisor observed that even a minor structural change, such as updating the grading scale, would probably require reframing the entire course and registering a totally new one. This example shows that even small improvements carry a heavy administrative burden. Consequently, these management spots hide from external participants too. The external stakeholder stated that the entire course setup remained confusing to him from start to finish. Across both cases, the core problem is not that any members forgot to follow specific routines, but that the university system itself lacks a flexible administrative foundation designed to make all information visible to all actors, handle everyday complications and adapt to feedback over time.

T5. Ambiguous mutual value and unframed deliverables

While previous tensions focus on daily operations, a more basic conflict regarding purpose and value of the project itself is identified. In the GDL case, this uncertainty appeared at the beginning of the project. The external stakeholder explained that company employees regularly questioned why they should participate and how could the collaboration help their regular jobs, noting that staff must justify their time to managers. This stakeholder also highlighted a conflict between looking at the bigger picture and focusing on immediate business results. This vagueness makes it difficult to find a balance, resulting in failing to provide a clear direction for students. This aligns perfectly with the student coordinator's comment on the lack of clear challenges from partners, which explains why Student 2 also felt the team had no clear starting point. Without official, co-defined goals, students felt forced to deliver satisfactory results to corporate partners to meet their own learning objectives under vague instructions, even though the course guidelines explicitly stated this was not their responsibility.

In the MF2088 case, this confusion emerged during the final prototyping stage. The external stakeholder found a common issue that it was difficult to turn student ideas into market-ready products. The value of the student solution stayed little above academic expectation but still under realistic application objectives. It was even noticed by external stakeholder during the prototyping phase when it was late to adjust big. This underlying misalignment about purpose possible causes and worsens the operational problems described in the previous sections.

5.4.2 From Systemic Tensions to Opportunity Areas

The five tensions identified above define five opportunity areas for redesigning the collaboration infrastructure of challenge-driven educational ecosystems. Rather than offering immediate solutions, these opportunity areas identify critical entry points where systemic interventions can generate long-term value across multiple stakeholder relationships and project phases. The opportunity areas are summarized in Figure 5.5 and outlined briefly here, since their translation into service design principles forms the starting point of Chapter 6.

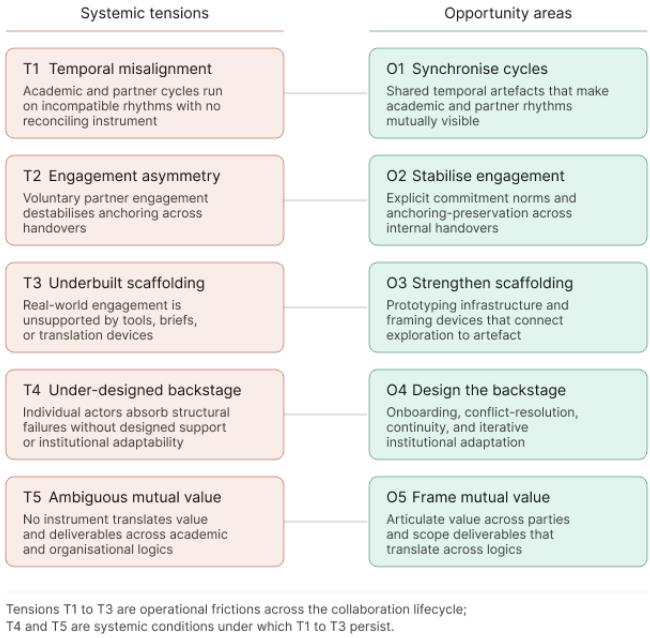


Figure 5.5 From systemic tensions to opportunity areas

Each opportunity area responds directly to one of the five tensions. O1 strengthens shared temporal coordination across academic and organizational rhythms (T1); O2 improves continuity, anchoring, and accountability across stakeholder participation (T2); O3 reinforces the collaborative support required for prototyping, access, framing, and real-world engagement (T3); O4 establishes stronger backstage support for coordination, onboarding, facilitation, and organizational continuity (T4); and O5 supports clearer articulation of mutual value, challenge framing, and translation between educational and organizational contexts (T5).

The five opportunity areas are interdependent rather than isolated, addressing the temporal and relational, infrastructural, backstage, and value-framing conditions of collaboration tensions. Changes in one area tend to influence others through shared actors, sequential dependencies in the collaboration journey, and feedback loops between stakeholder engagement and perceived value creation. Together, they emerge from the synthesis of stakeholder mapping, journeys, and service blueprint analysis developed throughout the chapter, and enable a transition from systemic diagnosis toward systemic intervention.

The opportunity areas show where need to be changed, but they do not show what the new collaboration tools or setups should look like. Chapter 6 fixes this problem by developing a set of service design principles based on these opportunity areas, which are later introduced into proposed collaboration infrastructure intervention.

6 Service Design Implication and Collaboration Infrastructure Redesign

6.1 From Opportunity Areas to Service Design Principles

Design principles connect user insights to technical implementation, defining systemic qualities independent of technological constraints[12]. Although Chapter 5 maps out where systemic change is needed, it does not define the concrete design principles required for the new collaboration infrastructure. This section addresses that gap by using the opportunity areas to develop service design principles. Instead of listing technical specifications, these principles define the core qualities needed to guide future work, turning broad findings into concrete design directions.

The mapping from five opportunity areas to four principles is not one-to-one. Opportunities O1 to O4 each correspond directly to a principle. Instead of forming a distinct principle, Opportunity O5 functions as an underlying condition that guides the application of the other four principles. This is consistent with the analytical claim, established in Chapter 5, that the opportunity areas are interdependent rather than isolated, and that value framing constitutes the qualitative base through which the other dimensions become meaningful to stakeholders. This translation is visualized in Figure 6.1.

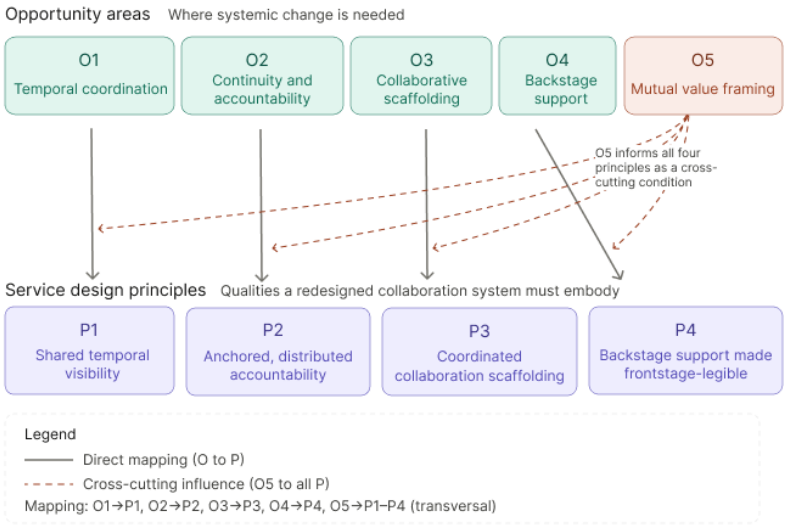


Figure 6.1 Opportunity areas (O1–O5) to service design principles (P1–P4)

1. **Shared Temporal Visibility** Stakeholders must operate within a mutually transparent framework that aligns academic milestones, organizational timelines, and partner availability throughout the project lifecycle. This principle directly addresses the temporal fragmentation identified in Chapter 5, reducing the coordination uncertainty caused by incompatible academic and corporate schedules.
2. **Anchored and Distributed Accountability** Roles, mutual expectations, and project contributions must be explicitly defined and remain accessible beyond personnel changes. Commitment should not rely on informal memory or the continued presence of a single individual. This principle responds to the fragility of organizational anchoring, mitigating the instability caused by shifting partner contacts and implicit responsibility structures.
3. **Coordinated Collaboration Support** The collaboration environment must provide structured, built-in support for onboarding, challenge framing, resource access, and exploratory design work. Essential context and participation expectations should be clear from the outset,

coordinated through a shared support framework rather than left to ad-hoc negotiation. This principle directly addresses the infrastructural gaps identified around vision alignment, interdisciplinary translation, and real-world access.

4. **Backstage Support Made Frontstage Legible** The coordination labor required to sustain a partnership must be structurally supported and visible to all participants. Making this backstage work transparent builds trust and ensures project continuity without overloading external partners with daily operational complexity. This principle responds to the invisible coordination labor observed across the case studies, including onboarding, facilitation, and conflict mediation.

While presented separately, these four principles are also deeply interconnected. Shared visibility supports accountability; explicit accountability strengthens continuity; continuity reduces administrative burdens; and backstage support enables the collaboration support. The cross-cutting role of mutual value framing reinforces this interdependence by shaping how each principle is interpreted in practice: what counts as a legitimate milestone, who is accountable for what, which challenges are worth to be supported, and whose coordination labor is recognized as valuable. Across all four dimensions, transparency, continuity, and adaptability emerge not as isolated functions but as the core qualities of a redesigned collaboration infrastructure.

6.2 Defining the Scope of Intervention

Before introducing the proposed framework, the boundaries of the redesign must be explicitly clarified to define the internal and external parameters of the intervention.

The primary object of this redesign is the collaborative infrastructure supporting challenge-driven educational projects. This includes the coordination mechanisms, onboarding structures, continuity practices, and information flows through which students, coordinators, supervisors, and external partners interact. Therefore, the intervention focuses strictly on the structural conditions that shape collaboration dynamics, rather than on the specific educational content of individual labs or courses. Accordingly, the redesign does not target curriculum structures, grading criteria, institutional governance, or the underlying pedagogical philosophy of challenge-driven education. These elements are treated as fixed contextual conditions within which the new framework must operate; the intervention is positioned not as a redesign of challenge-driven education itself, but as a redesign of the collaborative framework that enables it.

The primary actors within the scope of this intervention are student teams, course coordinators, and contact persons from partner organizations. Adjacent actors, including such as university administration, senior corporate executives, and project end-users, are recognized as part of the broader ecosystem but are not treated as direct users of the proposed tools. Similarly, institutional funding structures and macro-level policy reforms fall outside the boundaries of this study.

Within these defined boundaries, the proposed framework prioritizes four core functions: temporal coordination, expectation alignment, continuity support, and contribution visibility. The Collaboration Canvas is designed as an institutional support framework for university-industry deployment, rather than a standalone commercial digital product.

6.3 The Need for Integrated Collaborative Frameworks

Instead of relying on temporary solutions like individual onboarding sessions or facilitation guidelines, this study establishes an integrated framework to systematically resolve collaboration conflicts. This shift is rooted in the empirical findings of Chapter 5. The data shows that tensions across both cases were not random communication breakdowns or individual responsibility, but organizational vulnerabilities that span across time, stakeholder networks, and institutional

boundaries. For this reason, Problems like fragmented information sharing and low continuity are not accidental operational errors. Instead, they are structural flaws within the collaborative foundation, meaning that short-term adjustments cannot fix them in the long term.

Therefore, the proposed solution focuses on establishing a shared foundation to support continuity, coordination, and visibility throughout the entire project lifecycle.

This unified approach was prioritized and finally determined for three reasons. First, the empirical findings showed that fragmented coordination and information asymmetry affected all identified opportunity areas. A shared framework offers strong systemic value by resolving multiple tensions at once. Second, stakeholder engagement in challenge-driven collaborations is often fragile. Managing several disconnected tools would create unnecessary friction for external partners, whose participation is already limited by competing organizational priorities. Third, the lack of a structured knowledge repository caused recurring friction in both cases. A unified system ensures that onboarding materials, documentation, and project history are preserved across semesters and personnel turnover.

The proposed framework is not a technological fix for isolated operational errors. Instead, it redefines how collaboration visibility, continuity, and coordination are structured within challenge-driven educational models.

6.4 The Collaboration Canvas as a Systemic Framework

This section presents the proposed collaboration framework developed in response to the systemic tensions. Rather than introducing a standalone digital platform, this solution serves as a shared framework that translates the service design principles into practical tools. The developed framework is named *Collaboration Canvas*, which functions as a shared coordination layer where stakeholders could use to align expectations, document activities, and maintain project visibility

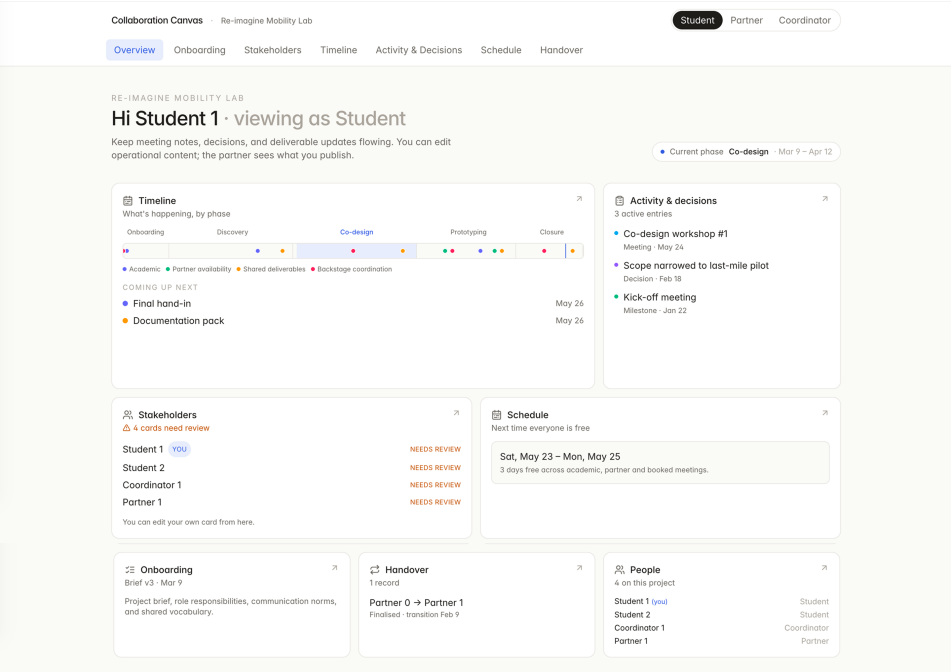


Figure 6.2 Overall Collaboration Canvas architecture and stakeholder interaction structure

throughout its lifecycle.

The Collaboration Canvas is structured as a timeline-based environment acting as the primary shared touchpoint. It does not replace existing communication channels like email or online video

conferences. Instead, it provides a persistent baseline that stabilizes coordination across fragmented academic and organizational workflows.

To balance the workload, the canvas implements differentiated participation structures. Students manage the majority of operational updates and project documentation, external partners provide lightweight inputs regarding availability and feedback, while coordinators handle backstage onboarding layers. This structure redistributes visibility without overburdening any single participant.

6.4.1 Timeline Backbone for Shared Temporal Alignment

The timeline backbone adapts the logic of journey mapping from an analytical tool into an active, multi-actor coordination structure.

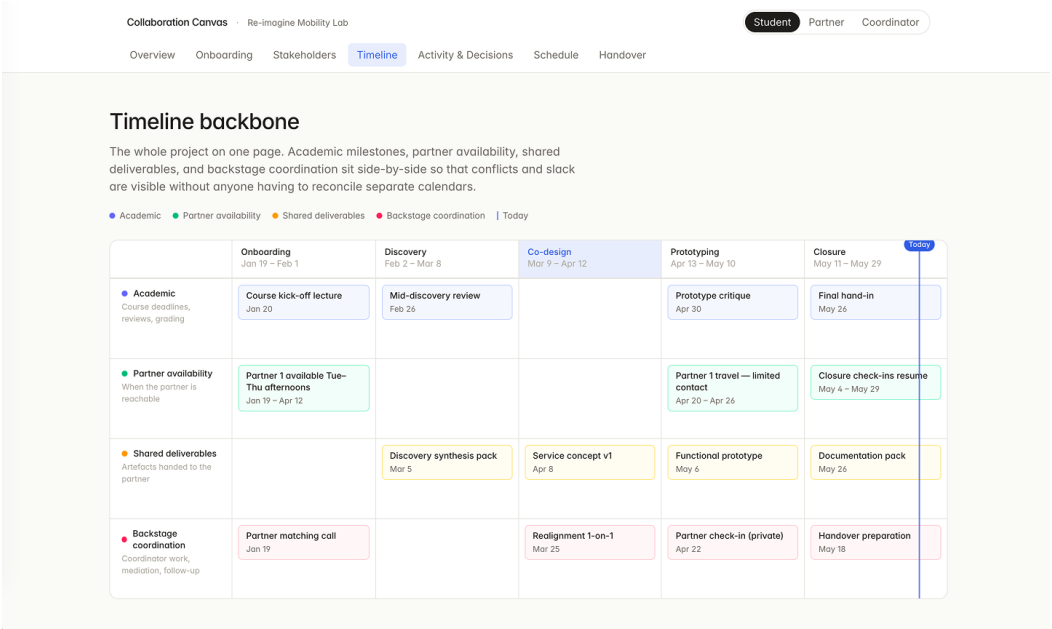


Figure 6.3 Timeline backbone supporting shared temporal visibility across stakeholder groups

The project lifecycle is organized into five phases: onboarding, discovery, co-design, prototyping, and closure. Within each phase, parallel lanes visualize academic milestones, stakeholder availability, shared deliverables, and backstage coordination. Rather than functioning purely as a static schedule, the timeline synchronizes collaboration rhythms across participants who operate under different organizational timelines. This live infrastructure externalizes temporal dependencies so that they would not remain fragmented across separate communication channels.

6.4.2 Stakeholder Value-Contribution Map

The stakeholder value-contribution map extends traditional stakeholder mapping by embedding mutual expectations and defined responsibilities directly into the tool.

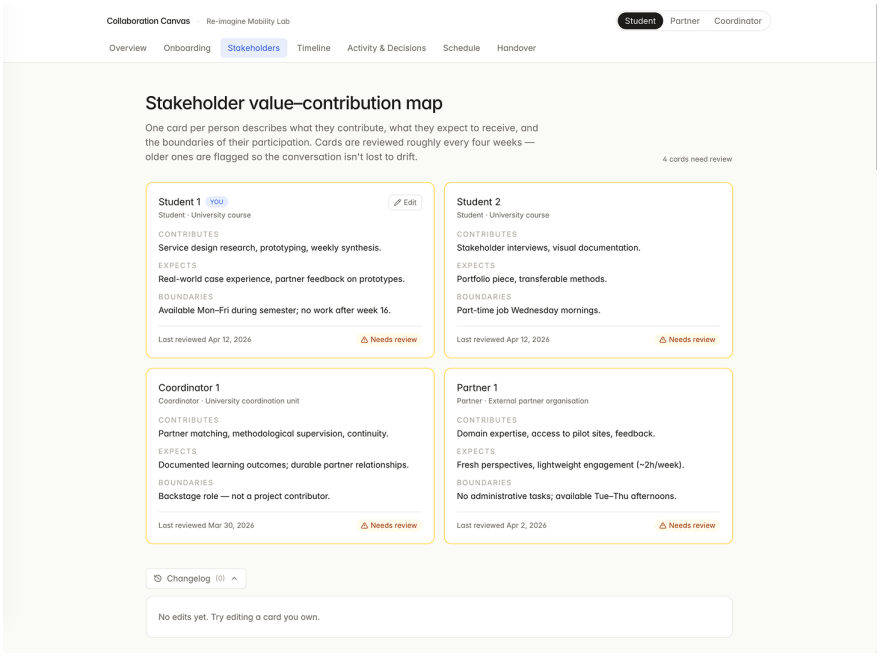


Figure 6.4 Stakeholder value–contribution structure supporting expectation alignment and accountability

Rather than serving as a static diagnostic diagram, the map is a dynamic resource used throughout the project lifecycle. Stakeholders use it to jointly clarify expected outcomes, available resources, and boundaries of participation. By making these commitments transparent early on, the component secures accountability beyond informal discussions. It directly addresses the ambiguity identified in Chapter 5 regarding stakeholder motivations and mutual expectations.

6.4.3 Activity and Decision Log

The activity and decision log provides a lightweight, shared project history to secure continuity.

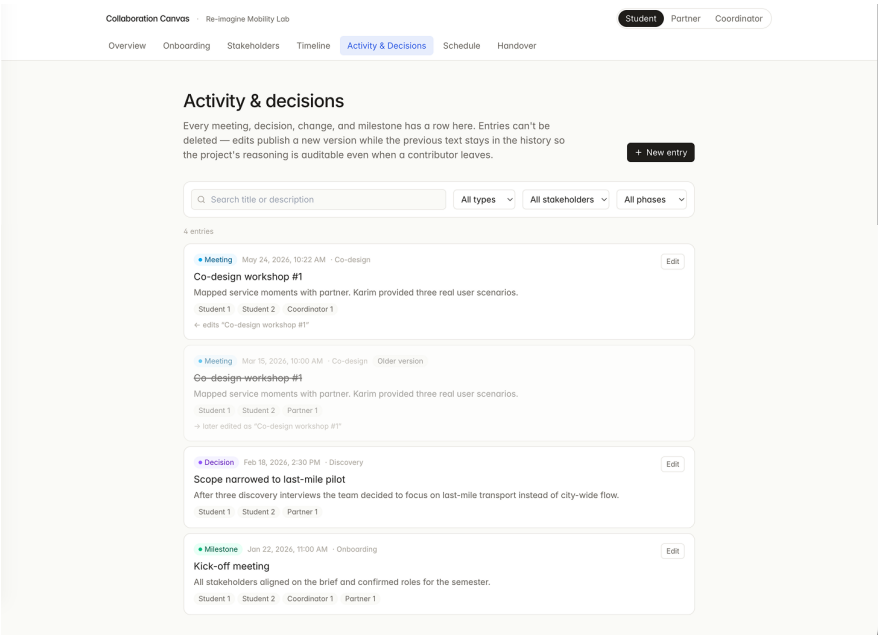


Figure 6.5 Activity and decision log supporting continuity and collaboration memory

Meetings, decisions, and project changes are documented chronologically. Entries are categorized and time-stamped for easy retrieval during onboarding and handover transitions. The goal is not administrative micromanagement, but the preservation of institutional knowledge. By externalizing project history into a shared artifact, the canvas reduces the project's reliance on fragmented personal messages and memory, ensuring a reliable records history across different project phases.

6.4.4 Onboarding and Expectation Alignment Zone

The onboarding zone consolidates essential project materials into a single, continuously accessible environment.

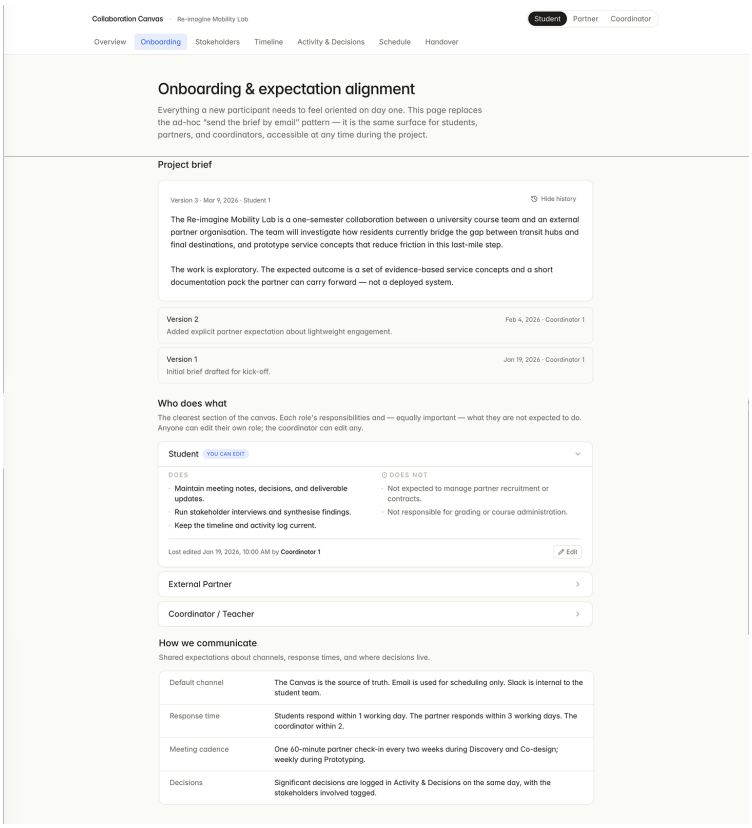


Figure 6.6 Onboarding and expectation-alignment layer consolidating shared project

Project briefs, timelines, role descriptions, and communication norms are maintained within this central structure. Version history is preserved to minimize confusion when project scopes evolve. This zone reframes onboarding not as a single orientation event, but as a permanent reference layer. It supports collaboration continuity while reducing the repetitive backstage facilitation work typically required from coordinators and supervisors.

6.4.5 Continuity and handover record

The continuity and handover record mitigates the disruption caused by staff turnover and changing stakeholder participation.

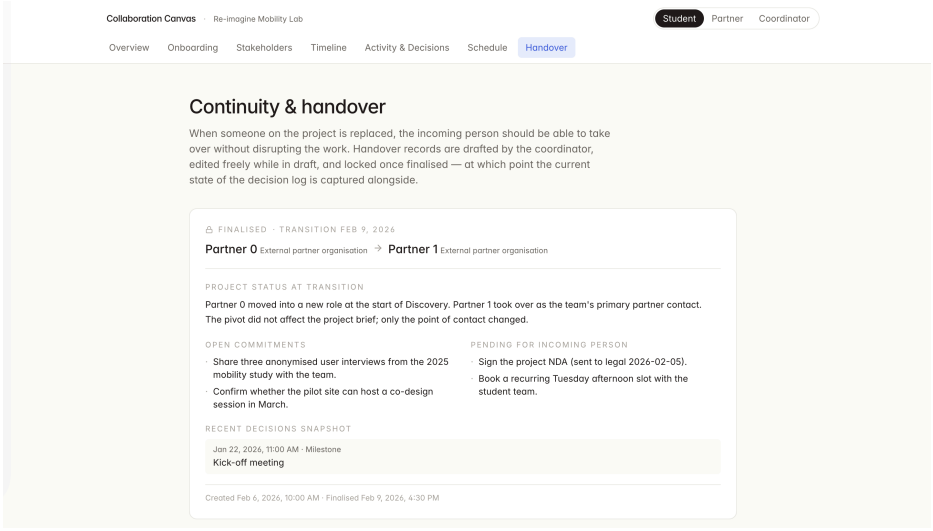


Figure 6.7 Continuity and handover structure supporting persistent stakeholder anchoring

When personnel changes occur, this component aggregates relevant project history, past decisions, and active commitments into a structured summary. Incoming participants gain immediate context regarding the project status without requiring extensive re-briefing from student teams. This tool transforms project continuity from an individual employee responsibility into a built-in system function, directly protecting the project against the organizational instability noted in the case studies.

6.4.6 Schedule overlay and coordination visibility

The schedule overlay visualizes the intersection between academic timelines, shared milestones, and external stakeholder availability.

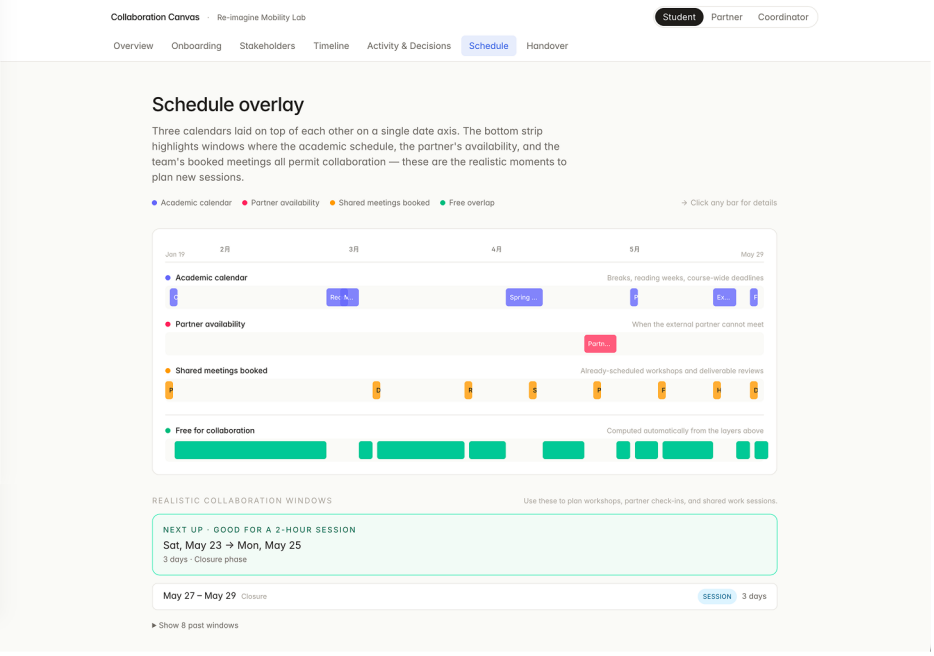


Figure 6.8 Schedule overlay supporting coordination visibility across academic and organizational timelines

Instead of trying to replace external corporate calendars, the overlay surfaces realistic collaboration windows to reduce scheduling friction. It clarifies the basic time constraints and conditions required for active engagement, making temporal dependencies visible across different organizational systems.

6.4.7 Role-differentiated access and backstage visibility

This component implements differentiated access structures to balance system transparency with stakeholder workload.

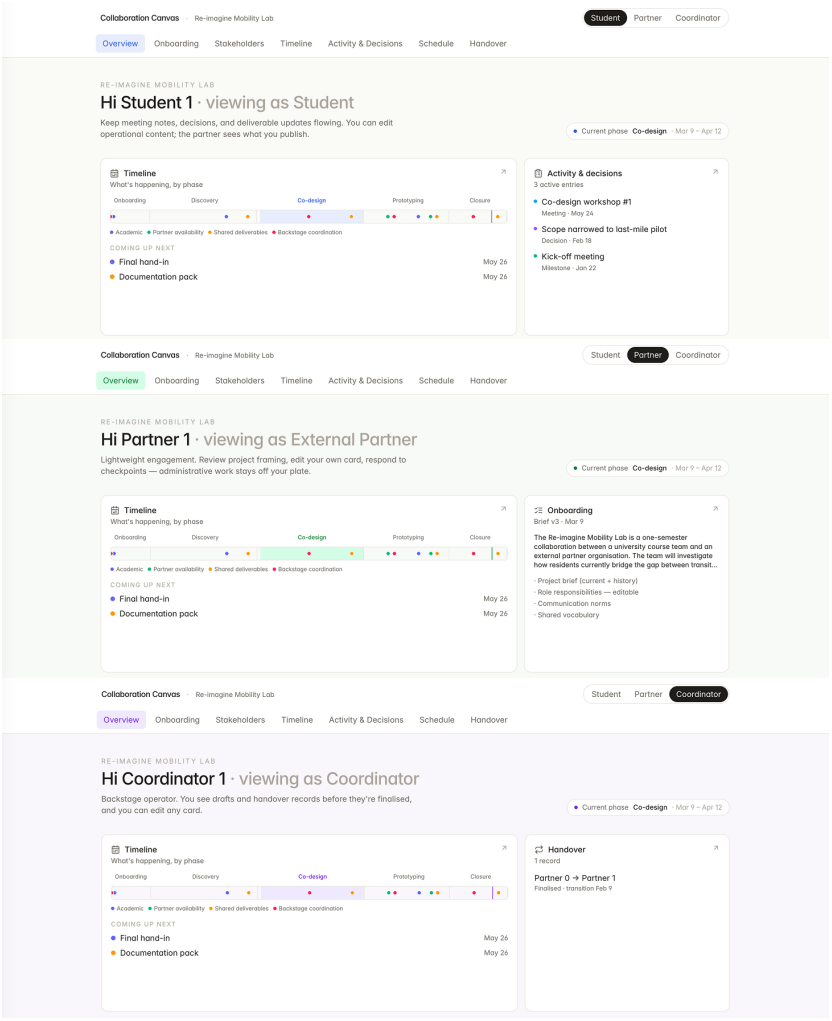


Figure 6.9 Role-differentiated access structure and backstage coordination visibility

Students maintain the majority of daily documentation and updates, reflecting their central role in project execution. External partners provide lightweight, structured feedback rather than handling administrative tasks, while supervisors retain access to backstage monitoring layers for timely intervention. This division of labor ensures that internal scheduling and administrative tasks remain well-organized, while protecting external partners from the confusing details of day-to-day project management.

6.5 Expected Systemic and Experiential Shift

The primary objective of the proposed framework is to improve both the systemic conditions and the stakeholder experience within challenge-driven education. While long-term empirical validation

of the framework falls outside the immediate scope of this thesis, distinct systemic and experiential shifts can be projected based on the empirical patterns identified in Chapter 5. To illustrate how the Collaboration Canvas transforms the collaborative environment, Table 6-1 contrasts the identified limitations of the current state with the expected outcomes of the redesigned framework.

Table 6-1 Systemic and experiential shifts from current to redesigned state

Dynamic / Dimension	Current State	Redesigned State
Collaboration Memory	Fragmented across personal notes, individual emails, and ad-hoc meetings, leading to information gaps.	Preserved through a persistent, shared coordination foundation accessible to all actors.
Expectation Alignment	Remains partially implicit; expectations are often revisited only after coordination breakdowns occur.	Continuously visible, tracked, and adjusted throughout the entire project lifecycle.
Backstage Labor	Critical coordination work by supervisors remains invisible, hiding the true effort required to sustain partnerships.	Clear procedures, all parties remain informed, fostering collaboration without creating extra administrative tasks for the staff."
Stakeholder Engagement	Fragile and vulnerable to staff turnover, relying heavily on informal knowledge transfer.	Systematically secured through built-in handover mechanisms independent of individual personnel changes.

Instead of viewing teamwork as a series of disconnected meetings, the updated project structure helps partners directly apply the students' initial research to their daily operations. By standardizing how expectations, contributions, and deliverables are framed, the Collaboration Canvas moves beyond individual stakeholder satisfaction. It directly restructures the foundational conditions through which cross-institutional collaboration is coordinated, sustained, and made meaningful.

7 Discussion

7.1 Discussion of Findings in Relation to Existing Literature

Collaboration challenges observed in GDL and MF2088 should not be understood only as isolated issues of communication, motivation, or stakeholder engagement. Instead, they point to deeper infrastructural tensions in how challenge-driven educational collaborations are organized, supported, and sustained over time. Existing literature on challenge-driven education has already identified the importance and difficulty of external stakeholder involvement, especially in relation to role ambiguity, varying commitment, and misaligned expectations [10], [11], [13]. This study extends those discussions by pointing out that such difficulties are not only the result of individual stakeholder behavior but are structurally shaped by the absence of systemic infrastructures designed for multi-stakeholder alignment, coordination, and lifecycle continuity.

A central finding concerns the fragility of stakeholder engagement across the collaboration lifecycle. In both cases, engagement depended heavily on individual initiative: the student coordinator in GDL, the teacher supervisor in MF2088, and specific contact persons within partner organizations. When these individuals were committed, collaboration could become meaningful and productive. However, when roles shifted, when feedback arrived late, or when partners lacked internal anchoring, the collaboration became unstable. This indicates that stakeholder engagement in challenge-driven education is not only a matter of inviting external actors into the process [10]. It also requires mechanisms that help actors remain connected, informed, and meaningfully involved across different phases of the project [11], [13].

Temporal misalignment further illustrates this infrastructural problem. Academic projects operate through fixed semester schedules, deadlines, and assessment structures, while external organizations follow business cycles, internal decision-making processes, changing agenda and resource priorities [10], [11]. In GDL, this was visible in late stakeholder feedback, unclear challenge framing at the beginning of the semester, and weak continuity between project cycles. In MF2088, the temporal issue appeared in delayed user access, partner reprioritization, and late increases in company engagement. These findings show that timing issues are not only practical problems of scheduling. They directly influence whether different stakeholders can align their expectations, contribute reasonably and effectively, and create value together [11], [35].

The service blueprint revealed that many visible collaboration breakdowns were caused by invisible backstage work that was under-supported. Students and partners mainly encountered frontstage activities such as meetings, workshops, presentations, and feedback sessions [39], [40]. However, the quality of these visible interactions depended on backstage activities such as recruitment, expectation translation, stakeholder anchoring, conflict mediation, handover, and follow-up [40]. In both cases, due to rigid institutional structure, the backstage work was heavily absorbed by individual coordinators and teachers acting as ad-hoc buffers. This finding strengthens the relevance of a service design perspective, which demonstrates that improving the visible stakeholder experience requires transforming these invisible, exhausting personal efforts into organized, institutionalized support [12], [40], [41].

The study also highlights a core tension in challenge-driven education. There should be a balance between open-ended exploration and practical applicability [13]. Wicked problems need to remain broad enough to support systemic learning, but external partners also need outcomes that are concrete enough to justify their participation [10], [13]. In GDL, students struggled with broad and under-defined challenges, while the external stakeholder questioned how exploratory outcomes could become meaningful for the organization. In MF2088, students produced more concrete concepts, but the transition from course output to real implementation remained difficult. This suggests that the value of challenge-driven collaboration cannot be assumed to emerge naturally

from openness [30], [34], [35]. It needs to be actively framed, negotiated, and translated throughout the project.

Taken together, the central challenge is not simply how to increase stakeholder participation, but how to design a collaboration infrastructure that can support participation over time. From this perspective, alignment tools, role clarification, value mapping, handover mechanisms, and backstage coordination should not be treated as administrative additions [12], [40], [41]. They are core components that drive meaningful collaboration in challenge-driven educational ecosystems. This reframing provides the basis for the contribution discussed in the following section.

7.2 Contribution to research on challenge-driven education and stakeholder engagement

This study contributes to research on challenge-driven education, stakeholder collaboration, and service design by shifting the analytical focus from stakeholder engagement as a participation issue to collaboration infrastructure as a design condition. Previous research has emphasized the pedagogical value of challenge-driven education, including interdisciplinary learning, engagement with real-world problems, and collaboration with external stakeholders [13], [21], [22], [23]. However, less attention has been paid to how the collaboration itself is structured, maintained, and experienced across the full project lifecycle. By adopting a service ecosystem perspective, this study shows that challenge-driven collaboration can be understood as a multi-actor service system in which value depends on continuous alignment, coordination, and expectation translation across students, coordinators, teachers, and external partners [30], [31], [34], [35].

The main theoretical contribution of the study is the reframing of recurring collaboration difficulties as infrastructural rather than personal or operational problems. Issues such as unstable engagement, unclear expectations, delayed feedback, and weak continuity are often visible at the level of meetings, communication, or stakeholder motivation [10], [11], [13]. This study shows that these issues are better understood as results of missing structures for role clarification, temporal coordination, mutual value articulation, backstage support and handover. In other words, stakeholder participation cannot rely only on individual commitment or informal communication. It needs clear structures that help different actors stay aligned and involved throughout the project, which provide support before, during, and after visible collaboration moments.

Another contribution concerns the role of service design methods within collaborative ecosystems. In the study, stakeholder mapping, journey mapping, and service blueprint gradually evolved from analytical tools into components of the collaborative infrastructure [39], [40]. This suggests that service design methods can move beyond diagnosis and become infrastructural devices. They can help make roles, expectations, timelines, dependencies, and backstage work more visible and negotiable among stakeholders. Moreover, apart from treating service design tools as descriptive mapping techniques, this study demonstrates their value as analytical mechanisms for identifying systemic leverage points in challenge-driven collaboration. By tracing recurring pain points and coordination breakdowns across multiple stakeholder journeys, the study shows how service design can support system-level interventions that strengthen the collaboration infrastructure.

Therefore, the contribution of this thesis is not simply the identification of collaboration problems in GDL and MF2088. It lies in proposing a conceptual shift: challenge-driven educational collaboration should be designed as an infrastructure-supported service ecosystem. This perspective connects pedagogical ambitions with the practical conditions needed to sustain multi-stakeholder collaboration over time. It also provides a foundation for future research and practice that examines not only what students learn from external collaboration, but how the collaboration system itself can be designed to support continuity, balance, and mutual value creation.

7.3 Practical Implications for GDL and Challenge-Driven Courses

This study has several practical implications for GDL and other challenge-driven educational initiatives involving long-term collaboration between academia and external organizations.

Early alignment should become a more explicit part of the collaboration process. Before students begin working on a challenge, external partners, coordinators, and student teams need a shared understanding of why the collaboration exists, what each party expects, and what kind of outcome is realistic. Partner engagement also needs clearer role and value framing, so that partners understand both what they are expected to contribute and what value they may gain from the collaboration. At the same time, stronger handover and continuity mechanisms are needed to carry project outcomes, feedback, unresolved questions, and possible next steps beyond one student cycle. Backstage coordination work, such as recruitment, expectation translation, communication support, and stakeholder follow-up, should be recognized as a necessary part of the collaboration rather than informal extra work. Collaboration does not become meaningful simply because people are brought into the same room. It becomes meaningful when everyone involved put an effort in and shares responsibility, as well as when the conditions around them help them understand, contribute, and continue.

The proposed Collaboration Canvas can be understood as one possible practical response to these needs. It does not replace existing teaching activities or stakeholder meetings, but provides a shared structure for making roles, expectations, timelines, value exchanges, and handover information more visible. In other words, the practical implication of the study is that challenge-driven collaboration should be managed not only as a course or project, but as a service system that requires continuous coordination and visible infrastructure. The implications for actors within the ecosystem are the importance of reaching a consensus on value and expectations, knowing that the collaboration concerns everyone and communication in need of support from all stakeholders.

7.4 Methodological reflections

This study combines qualitative research using semi-structured interviews with service design methods including stakeholder mapping, journey mapping and service blueprint. These methods proved to be valuable not only as visualization tools, but also analytical approaches that help reveal different layers of the collaboration process, even the hidden backstage supports. As described in Chapter 4, each tool performed as expected. Stakeholder maps visualize the role dependencies separately in each case. Journey maps traced how tensions evolved across different project phase from start to end instead of appearing as isolated events. Service blueprint was extremely useful to identify backstage coordination activities that were usually invisible. Besides, the iterative process between interviews, thematic synthesis and service design visualization also facilitated a more systematic interpretation and analysis of empirical data. This combined approach connects individual stakeholder perspectives rather than analyzing them in isolation. When similar conflicts appear repeatedly, the data are grouped together to explore how these issues simultaneously affect the broader organization, the user journey, and the shared tools.

However, there are still certain limitations of this methodological approach. As service design methods are inherently interpretive, translating empirical data into maps and blueprints inevitably involves researcher subjectivity. Therefore, the resulting models represent analytical interpretations rather than objective realities. Nevertheless, the study's validity was strengthened by triangulating feedback across multiple stakeholder groups and conducting a comparative analysis between the two empirical cases. This rigorous process ensured the consistency of the identified tensions and systemic gaps.

AI Tool Usage Reflection

The research made use of several AI-assisted tools to support different stages of the project, including literature exploration, interview transcription, data organization, writing support, and conceptual development. Rather than replacing the researchers' analytical work, these tools primarily served as productivity aids with benefits and limitations as shown in table 7-1.

Table 7-1 Overview of AI Tools, Their Applications, Advantages, and Limitations

AI Tools	Usage in Study	Advantages	Limitations
Granola	Automatic transcription and note generation for interviews	Significantly reduced the time required for transcription and facilitated rapid review of interview content	Transcription errors may occur, expressions specific to the context may be omitted, or technical terms may be misunderstood. All transcribed text must be manually cross-checked against the original recording being used
Gemini and ChatGPT	Generating ideas, language refinement and cross-checking explanations	Offered diverse viewpoints and useful suggestions during ideation and conceptual development, improved writing and assisted in reorganizing complex ideas into clearer academic language	Ideas might deviate from the topic or too hard to be realized. Suggestions could change nuance or introduce interpretations not fully supported by the empirical data, requiring careful human review
Claude	Development assistance for building the interactive Collaboration Canvas prototype	Helped turn the service design concept into a working interactive prototype, generated and refined the prototype code, and allowed fast iteration on layout and the embedded tools without separate development resources	Generated code needed testing and debugging, and was checked and corrected by the author before use
Primo Research Assistant of KTH library and Perplexity	Literature discovery, information retrieval and identifying relevant references	Enabled efficient exploration of topics and quick access to supporting sources, helping to accelerate background research	Retrieved information still required independent evaluation for credibility, relevance, and accuracy, and references needed to be verified before inclusion in the thesis

The use of AI tools greatly improved the efficiency of the research process by reducing repetitive tasks and supporting drafting, information retrieval, and organization. However, these benefits were accompanied by an increased need for verification and critical evaluation. AI-generated outputs

were never treated as authoritative and were systematically reviewed before being incorporated into the study. Especially when using automated transcription tools such as Granola. Since transcription inaccuracies could affect qualitative coding and interpretation, the researchers returned to the original recordings whenever ambiguity or discrepancies were identified. Similarly, suggestions generated by large language models were used only as writing or ideation support rather than as analytical evidence.

As AI tools becomes increasingly integrated into academic research, its responsible use requires not only utilizing its efficiency but also maintaining transparency, critical thinking, and rigorous validation against original data sources. The increasing usage of AI tools does not reduce the need of researcher involvement but rather shift researcher's roles from performing routine tasks to critically evaluating, validating, and contextualizing AI-generated results.

7.5 Limitations and Future Research

Overall, this study also has several limitations. First, the empirical data in this study is based only on a limited number of interviews conducted in two empirical cases. Although these cases provide complementary insights, the findings may not fully reflect the diversity of all challenge-driven education situations. Furthermore, this study primarily focuses on stakeholders directly involved in project coordination and implementation, while broader institutional actors within the collaborative ecosystem such as university leadership or corporate management layer are not included in the scope of the research. At the same time, the proposed canvas remains at the conceptual stage. Although it is grounded in empirical research, it has not yet experienced long-term evaluation in real projects. Therefore, its practical utility remains to be assessed.

These limitations point to a clear direction for future research. A key next step is to validate the canvas in practice and iteratively optimize it. This can be made in the form of stakeholder workshops or academic seminars, enabling the tool to continuously evolve based on realistic feedback. In addition to tool validation, further research could explore how these collaborative infrastructures adapt to different institutional contexts, disciplinary traditions, or organizational cultures. Moreover, exploring new technologies can help address these challenges. For example, investigating on how AI-supported coordination tools or adaptive guidance systems might support or reshape these educational ecosystems over the long term could be potential exploratory direction.

8 Conclusions

This study explored how stakeholder collaboration is experienced and coordinated within challenge-driven educational ecosystems through the cases of GDL and MF2088. The findings revealed that recurring collaboration breakdowns were not isolated operational issues, but interconnected tensions related to temporal misalignment, fragile stakeholder anchoring, weak continuity structures, and invisible coordination work. From a service design perspective, the study further demonstrated how these tensions could be reframed as infrastructural conditions rather than interpersonal or pedagogical challenges. In response, the thesis proposed the Collaboration Canvas as an infrastructure-level intervention approach to support continuity, visibility, and coordination across multi-stakeholder collaboration ecosystems.

As for research questions, conclusions could be drawn as follows.

RQ1. Different stakeholders experienced the collaboration as meaningful but unevenly supported. Students valued the opportunity to engage with real-world complexity, but often struggled with unclear challenge framing, limited guidance, and uncertainty around stakeholder expectations. Coordinator and supervisor experienced the collaboration as dependent on significant backstage work, including recruitment, onboarding, expectation management, conflict mediation, and continuity maintenance. External stakeholders recognized the potential value of student collaboration, but also expressed uncertainty around time commitment, organizational relevance, and how broad exploratory work could translate into actionable outcomes.

RQ2. Five systemic tensions occur within the collaboration lifecycle: temporal misalignment (T1), asymmetric engagement and fragile anchoring (T2), underbuilt support for prototyping and real-world access (T3), under-designed backstage support (T4), and ambiguous mutual value (T5). These are ongoing issues built into the system itself, not just temporary errors at specific steps of the process.

RQ3. The collaboration infrastructure can be redesigned around four service design principles: shared temporal visibility, anchored accountability, coordinated support, and backstage support made frontstage-legible. The Collaboration Canvas operationalizes these principles as a shared coordination layer.

By shifting the analytical focus from surface-level engagement to deep structural support, this study lays the groundwork for future institutional design. The theoretical and practical framework established in this study opens up a critical way for validating collaborative infrastructure within longer teaching cycles and multidisciplinary contexts. As the educational ecosystem continues to evolve, exploring how adaptive technologies and automated coordination tools can integrate with these human-centered infrastructures remains crucial for fostering more resilient and sustainable industry-academia partnerships.

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Appendix A: Interview Guide – External Stakeholders

Interview guide A — External Stakeholders

> **Participants:** Partners from Companies / Public Organizations
> **Duration:** 45-50 minutes

Block 1 — Motivation & Expectation (Pre-Engagement) (10 mins)

`[TC-M]` → `JM (Phase 1: Pre-service)` → `SM`

1. How did you first become involved with GDL/this specific course?
   - `▶ Probe:` Through what channel (email, seminar, colleague recommendation) did you learn about this project?
   - `▶ Probe:` Was it your idea to join, or were you invited/assigned by someone else?
2. Before the project started, what were your main motivations? What were you hoping to achieve for your organization?
   - `▶ Probe:` Were these expectations clearly discussed with the university, or did you assume them yourself?
3. How do you view the student team in relation to your company?
   - `▶ Probe:` What are the differences in resource allocation (data, access permissions) compared to your internal teams? Are students given the same level of access and resources as your internal teams, or are there restrictions?
4. How was the onboarding process? (e.g., what documents, meetings, or agreements happened before you met the students?)
   - `▶ Probe:` Was there anything that felt unclear or missing during the onboarding?

Block 2 — The Collaboration Journey (Mid-Engagement) (20 mins)

`[TC-T]` `[TC-R]` `[TC-P]` → `JM (Phase 2: During)` → `BP (Frontstage)`

1. Could you describe the collaboration process from kick-off to the final delivery. What were the key milestones or interactions?
    - `▶ Probe:` Who initiated most of the contact — you, the students, or the university coordinator?
2. To participate in this project, what "invisible" preparatory work did you need to do within your company? (For example: convincing your boss, applying for budget, communicating with legal about data security, etc.)
3. How did you communicate with the student team? (e.g., email, Zoom, physical meetings, workshops)
    - `▶ Probe:` Which channel worked best? Was there a format that felt too formal or too informal?
    - `▶ Probe:` How regularly did you communicate with each others? Did you satisfy with the frequency? Feel reasonable/too few/too many physical
4. How would you describe your role throughout the project? Were you more of a "client," a "mentor," or a "co-creator"?
    - `▶ Probe:` Was this the role you *wanted* to play? Did anyone clarify your role in advance?
    - How does the company contact person report internally? Do they need to convince superiors to maintain investment in GDL?
5. **How do your colleagues or superiors view your participation in this "university project"? Do they see it as a source of innovation, or as an additional burden?**
6. Throughout the collaboration, did your organization's work pace ever clash with the university's timeline (e.g., semester deadlines, exam periods)?
    - `▶ Probe 1:` Can you describe a specific instance where this timing mismatch was particularly noticeable? What happened?

- ▶ Probe 2: How did this time misalignment affect your willingness to participate or the final outcome?
- ▶ Probe 3: If you could redesign the timeline, what adjustments would you make?

7. How did the project conclude? Can you walk us through the final phase — from the last deliverable to the handover or final presentation?
   - ▶ Probe: Was there a clear ending point, or did it feel like the collaboration just faded out?
   - ▶ Probe: How satisfied were you with how the project wrapped up? Was there anything left unresolved?

Block 3 — Pain Points & Breakdowns (10 mins)

`[TC-P]` → `JM (Emotional Low)` → `BP (Failure Points)` → `CD`

1. Is there a frustrating moment in this collaboration? What happened?
   - ▶ Probe: What was the main reason of it?
2. Did you ever feel there was a misalignment between what your organization needed and what the students were working on?
   a. If yes:
   - ▶ Probe 1: When did you first realize this misalignment? Was it early in the project or later?
   - ▶ Probe 2: What did you do when you noticed it? Did you communicate it to the students or the university?
   - ▶ Probe 3: Looking back, what do you think caused this misalignment? Was it unclear expectations, communication gaps, or something else?

Block 4 — Value & Outcomes (Post-Engagement) (10 mins)

`[TC-V]` → `JM (Phase 3: Post-service)` → `BP (Value Exchange)`

1. What was the actual outcome of the collaboration for your organization? (e.g., a report, a prototype, new ideas, a new recruit).
    - `▶ Probe:` Did you use or implement any part of the students' work? If not, why not?
2. Compared to what you expected at the beginning, how would you rate the value you received? Scale from 1-5, (1-Far below expectation, 3-met expectation, 5-far exceed expectation)
    - `▶ Probe:` What would possibly make the outcome more useful or actionable for you?
3. Would you be willing to participate in a similar collaboration again? What changes would make you more confident about participating again?

- `▶ Probe 1:` For example, in terms of communication methods, timeline coordination, role clarity, or support from the university?
- `▶ Probe 2:` Thinking back to the challenges you mentioned earlier，what specific mechanism or tool would have addressed those issues?

4. "Please recall a specific moment when you thought: 'This is great, this is exactly the collaboration we wanted!' What happened at that moment?" "Conversely, which moment made you feel: 'This might be a waste of time'? What specific communication or technical obstacles caused this?"

Block 5 — Future Vision (5 mins)

`[TC-F]` → `CD` → `BP (To-be)`

1. If we could design a new "onboarding kit" or a "collaboration playbook" for companies joining future GDL labs, what is the *one thing* it absolutely must include?
2. Is there a tool, platform, or structure from your professional world that you wish the university could adopt for managing this type of collaboration?

Closing (2 mins)

- Is there anything important about your experience that we haven't covered?

Appendix B: Interview Guide – Students

Interview guide B — Students

> **Participants:** Students who participated in GDL or MF2088 course
> **Duration:** 45–60 minutes

Block 1 — Expectations & Entry (Pre-Engagement) (8 mins)

`[TC-M]` `[TC-R]` → `JM (Phase 1)` → `SM`

1. Thinking back to the time before your very first contact with the company, how did you picture the collaboration happening? What kind of working relationship did you imagine you would have with them?
2. What were your personal expectations for this project? What did you hope to learn or gain from working with a company?
   - `▶ Probe:` Were these expectations mainly academic (e.g., learning new methods), professional (e.g., networking, real-world experience), or personal (e.g., solving a meaningful problem)?
3. How did the course prepare you for working with a company? Did you get any guidance or tools to help manage that relationship?
   - `▶ Probe:` Looking back, was there any information, tool, or support you wish had been available to you before or right at the start of the collaboration?

Block 2 — The Collaboration Journey (Mid-Engagement) (20 mins)

`[TC-T]` `[TC-P]` `[TC-S]` → `JM (Phase 2)` → `BP (Frontstage + Backstage)`

1. Can you describe the full journey of your collaboration with the company, from start to finish? What were the key meetings and critical moments?
   - `▶ Probe:` For example, think about the first meeting, mid-project check-ins, any major decisions or changes, and the final presentation. Feel free to

mark those moments on the timeline as we go. Which of these stood out to you most?

2. "Let's look at how communication actually worked day-to-day. Who usually started the contact each week — your team or the company?
3. Thinking about the whole project, who felt like they were driving the decisions most of the time — your team, the company contact, both equally, or someone else? How did that feel to you?
4. Did you ever feel the company's level of engagement changed during the project? (e.g., highly involved at first, then gradually disappearing — or vice versa)
    - `▶ Probe:` How did that shift in engagement affect your team's motivation and your final output?
5. When things got unclear or difficult with the company, who did you turn to for support? (teachers, course coordinators, peers)
    - `▶ Probe:` Was that support enough? Was it easy to access? What kind of support from the university or teachers?

Block 3 — Pain Points & Misalignments (10 mins)

`[TC-P]` → `JM (Emotional Low)` → `BP (Failure Points)`

1. Describe a specific moment when you and the company were clearly *not on the same page*. What happened, and how did you resolve it — or did you not resolve it?
2. How did you experience the balance between what the company seemed to need and what the course required from you? Were there times when those two pulled in different directions, and if so, how did you navigate that?
    - `▶ Probe:` How did you navigate that tension?
3. How did the collaboration actually end? can you introduce the final phase — the last deliverable, the final presentation, and what happened after.

- `▶ Probe:` Was there a clear ending, or did it just fade out? Was there any follow-up from the company? Did you hear whether they used your work?

4. After the project ended, what was the nature of your relationship with the company? Did you stay in touch, and if so, in what capacity?
    - `▶ Probe:` For example, did they offer you any follow-up opportunities, such as internships, job offers, or further collaboration? Or did the relationship end with the final presentation?

Block 4 — Outcomes & Reflection (Post-Engagement) (8 mins)

`[TC-V]` → `JM (Phase 3)` → `CD`

1. Looking back, did you feel you were treated more as students doing coursework or more as professional collaborators with the company? How did that feel, and does (or doesn't) the distinction matter to you? WHY?
2. What did you personally learn from this collaboration that you couldn't have learned from coursework alone?
   → About working with external stakeholders, navigating organizational dynamics, or about the actual challenge topic?

Block 5 — Future Design (5 mins)

`[TC-F]` → `CD`

1. If you could design one new tool, process, or rule for how the course manages student-company collaboration, how would it be?
2. If a friend was about to start a similar project, what's the one thing you'd remind them about, and the one thing you'd tell them to make the most of?

Closing (2 mins)

- At the very beginning of our conversation, you described your overall experience as [repeat their exact words]. Now that we've talked through the

whole journey in detail — would you still use the same words, or has anything shifted in how you see it?

- Anything else you'd like to add that you feel is important?
- May we contact you again if we need clarifications?

Appendix C: Interview Guide – University Coordinators

Interview guide C — University Coordinators

> **Participants:** GDL course teachers, course coordinators
> **Duration:** 50–60 minutes

Block 1 — System Vision & Setup (Pre-service) (10 mins)

`[TC-M]` `[TC-R]` → `BP (Backstage)` → `SM`

1. How do you define a "successful" collaboration in GDL or MF2088? What does it look like from your perspective?
2. How do you get in contact with external stakeholders? Is there a standardized process, or is it largely relationship-based and informal?
   - `▶ Probe:` Is there a written agreement with companies?
3. How do you currently align the expectations of companies with the academic requirements before the project starts?
   - `▶ Probe:` How do you help companies understand what to expect from student teams? Is there a briefing session for companies? What does it cover?

Block 2 — Managing the Service (Mid-service) (20 mins)

`[TC-T]` `[TC-S]` `[TC-P]` → `BP (Backstage + Support Processes)` → `JM`

1. Once a project is running, how do you monitor the quality of the collaboration between students and companies?
   - `▶ Probe:` Do you have regular check-ins with the companies? With the students? Who initiates them?
2. From your observations, at what phase of the project does collaboration most often break down? What are the most common causes?

3. Have you noticed differences between how large corporations vs. startups vs. public organizations engage with students?
    - `▶ Probe:` For example, do larger companies tend to have more structured processes, while startups are more flexible but less available? How do these differences affect the student experience?
4. What are the biggest "backstage" challenges for you as an organizer, like things the students and companies never see but that take significant effort to manage?
    - `▶ Probe:` For example, managing last-minute changes, dealing with unresponsive companies, balancing different stakeholder expectations, or handling conflicts that arise mid-project?

Block 3 — Service Failures & Recovery (10 mins)

`[TC-P]` → `BP (Failure Modes & Recovery)` → `CD`

1. Can you describe a specific case where a company collaboration went badly and what happened step by step, from warning signs to how it was resolved.
    - `▶ Probe:` What were the early warning signs that something wasn't working? At what point did you realize intervention was needed? Who was involved in resolving it, and what actions were taken?
2. Do you have a standardized process for handling situations where a collaboration fails, or is each case addressed individually?
3. Are there structural causes of failure that appear repeatedly, things that are built into the current design of the lab / course?
    - `▶ Probe:` For example, misaligned timelines, unclear roles, lack of onboarding for companies, or gaps in how challenges are scoped at the start?

Block 4 — Tools, Infrastructure & Future Design (15 mins)

`[TC-S]` `[TC-F]` `[TC-P]` → `BP (To-be: Backstage + Support)` → `CD`

1. How do you currently organize and facilitate the collaboration?

- ▶ Probe: Which of these tools are actively used by stakeholders, and which tend to be overlooked or unused?

2. Looking at the full (GDL ecosystem / MF2088 course project), which actor (student, company, teacher, coordinator) do you feel is currently least supported by the existing infrastructure?
   - ▶ Probe: Why do you think that is? What specific support or resources are they missing?
3. If you could redesign one aspect of how GDL operates，(whether it's how companies are onboarded, how students are supported during the project, or how the course structure is organized)，what would you most likely to change and why?
   - ▶ Probe: What specific problem would this change solve? Who would benefit most from it?
4. When evaluating student performance, do we also have a closed-loop mechanism for evaluating 'external partner performance'? If not, what are the barriers?
5. When information asymmetry occurs between the company and students, at which communication stage does this information typically get lost? What causes this?
6. Do you have suggestions that might improve the whole experience for all the stakeholders?

Closing (2 mins)

- Is there anything else you would like to add or share with us?
- May we follow up with you during our analysis phase?

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